DatecsPay card readers communication protocol

Version: 1.9

Change log

Version	Changes	Date
1.1	Added "Note" and "Note 2" descriptions in subcommand "0x01: RECEIVE DATA". Added "Note" descriptions in subcommands "0x01: START PURCHASE + GRATUITY / TIP" and "0x02: START PURCHASE + CASHBACK". Added tags 0x9F04, 0xDF7D, 0xDF7E and 0xDF7F in the "7. TAGS" section. Updated the document structure and added few other updated descriptions.	10.09.2020
1.2	Added section "10. TRANSACTION FLOW. Added "Note" description in section "11. EXAMPLE TRANSACTION".	11.09.2020
1.3	Added the following subcommands: 0x0B: DELETE BATCH 0x0C: CLEAR REVERSAL 0x1D: GET MENU INFO 0x1E: GET PUBLIC KEYS LIST 0x1F: GET SYMMETRIC KEYS LIST 0x20: EDIT COMMUNICATION PARAMETERS 0x21: KEYS COMMAND	01.02.2021
1.4	Added the following subcommand: 0x23: GET PASSWORDS	02.02.2021
1.5	0x23: GET PASSWORDS changed to 0x23: CHECK PASSWORD	05.02.2021
1.6	Added the following subcommands: 0x25: SELECT CHIP APPLICATION 0x26: GET CARD READER STATE Added the following subevent: 0x3F: SELECT CHIP APPLICATION	16.02.2021
1.7	Added the following sections: 12.CARD READER ERROR STRINGS 13.HOST ERROR STRINGS Updated section "6.Display result message"! Added tags DF8004 - DF8006 in "1. 0x01: TRANSACTION COMPLETE" Added subevent "1.0x82: USER INTERFACE" Added "Note" in "10. Transaction flow" → "Transaction loop"	12.01.2022
1.8	Added subcommand "4. 0x03: START PURCHASE + REFERENCE" Added subevent "3. 0x03: PRINT HANG TRANSACTION RECEIPT" Added "Note" in "13. 0x0A: START END OF DAY"	18.07.2022
1.9	Added tags 0x9F1E and 0xDF32 in section "7. TAGS" Added subcommand "0x27: GET TERMINAL TAGS"	08.11.2022

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1. Packet format

For convenience, we will use “card reader” or “pinpad” for indicating the DatecsPay card reader, and “ext device” for indicating the external device, which is communicating with the card reader (this can be a mobile device, tablet, PC and etc).

- **EXT DEVICE TO CARD READER:**

  '>' CMD 00 LH LL <DATA> CSUM

'>':	start paket (ASCII symbol '>')
CMD:	command number
LH:LL	length of DATA
CSUM:	xor of all bytes in the packet

  Example:
  3E 3D 00 00 01 00 02

- **CARD READER TO EXT DEVICE:**

  '>' 00 ST LH LL <DATA> CSUM

'>':	start paket (ASCII symbol '>')
ST:	status (see error codes)
LH:LL	length of DATA
CSUM:	xor of all bytes in the packet

  Example:
  3E 00 00 00 00 3E

The timeout between different bytes is 2 seconds.
Once the card reader receives a valid packet, it will return a response to the external device, where the field ST in the response will indicate if the command is executed successfully or if there is any error, which occurred. The external device can set timeout = 5 seconds for awaiting a response from the card reader after any kind of command was sent to the card reader.

Commands, which can be sent to the card reader:

- 0x3D – Borica command
- 0x40 – External internet command

Events, which can be received unexpectedly from the card reader:

- 0x0E – Borica event
- 0x0F – External internet event

Every command can contain multiple subcommands and every event can contain multiple subevents. All subcommands and subevents are packed in the <DATA> field – usually the first byte of the <DATA> field (DATA[0]).

2. 0x3D: Borica subcommands

Borica command contains multiple subcommands, which are associated with almost all activities, which are possible to be executed from the card reader, like for example loading configurations, starting transactions, getting information for already started or already finished transactions and etc.

1. 0x00: PING

DESCRIPTION:

Ping the card reader.

EXT DEVICE:

3D 00 LH LL [SUBCMD]

CARD READER:

00 ST LH LL

EXAMPLE:

(EXT DEVICE -> CARD READER) >> 3E 3D 00 00 01 00 02
(CARD READER -> EXT DEVICE) << 3E 00 00 00 00 3E

CLARIFICATION:

(EXT DEVICE -> CARD READER):

'>':	0x3E - Start packet.
CMD:	0x3D - Borica command.
00:	0x00 - Fixed value '00'.
LH LL:	0x00 0x01 - The data length is 1 byte.
SUBCMD:	0x00 - PING subcommand.
CSUM:	0x02 - Xor of all bytes (3E 3D 00 00 01 00).

(CARD READER -> EXT DEVICE):

'>':	0x3E - Start packet.
00:	0x00 - Fixed value '00'.
ST:	0x00 - No error (ERROR CODES).
LH LL :	0x00 0x00 - The data length is 0 bytes.
CSUM:	0x3E - Xor of all bytes (3E 00 00 00 00).

2. 0x01: TRANSACTION START

DESCRIPTION:

Start a transaction.

EXT DEVICE:

3D 00 LH LL [SUBCMD] <DATA: VAR>

<DATA>:

[Transaction type]:

[1..14] TRANSACTION TYPE
01 - PURCHASE
02 - PURCHASE + CASHBACK
03 - PURCHASE + REFERENCE
04 - CASH ADVANCE
05 - AUTHORIZATION
06 - PURCHASE + CODE
07 - VOID OF PURCHASE
08 - VOID OF CASH ADVANCE
09 - VOID OF AUTHORIZATION
10 - END OF DAY
11 - LOYALTY BALANCE
12 - LOYALTY SPEND
13 - VOID OF LOYALTY SPEND
14 - TEST CONNECTION
15 - TMS UPDATE

<Transaction parameters: VAR>:

Tags, encoded in TLV format, depending from the transaction type (For reference see "TLV STRUCTURE" section below):

BOR_CMD_TAG_AMOUNT	0x81	- [HEX BIG ENDIAN - 4 bytes]
BOR_CMD_TAG_CASHBACK	0x9F04	- [HEX BIG ENDIAN - 4 bytes]
BOR_CMD_TAG_RRN	0xDF01	- [STRING]
BOR_CMD_TAG_AUTH_ID	0xDF02	- [STRING]
BOR_CMD_TAG_REF	0xDF03	- [STRING]
BOR_CMD_TAG_TIP	0xDF63	- [HEX BIG ENDIAN - 4 bytes]

CARD READER:

00 ST LH LL

1. 0x01: START PURCHASE

NEEDED TAGS: 0x81

EXAMPLE:

(EXT DEVICE -> CARD READER) >> 3E 3D 00 00 08 01 01 81 04 00 00 00 32 BC
(CARD READER -> EXT DEVICE) << 3E 00 00 00 00 3E

CLARIFICATION:

(EXT DEVICE -> CARD READER):

'>':	0x3E - Start packet.
CMD:	0x3D - Borica command.
00:	0x00 - Fixed value '00'.
LH LL:	0x00 0x08 - The data length is 8 bytes.

SUBCMD: 0x01 - TRANSACTION START subcommand.
[TRANS TYPE]: 0x01 - HEX format, PURCHASE
<TRANS PARAMS: VAR>: '81 04 00 00 00 32' - AMOUNT (tag 0x81) = 50 (0.50)
CSUM: 0xBC - Xor of all bytes (3E 3D 00 ..... 00 32).

(CARD READER -> EXT DEVICE):
'>': 0x3E - Start packet.
00: 0x00 - Fixed value '00'.
ST: 0x00 - No error (ERROR CODES).
LH LL : 0x00 0x00 - The data length is 0 bytes.
CSUM: 0x3E - Xor of all bytes (3E 00 00 00 00).

2. 0x01: START PURCHASE + GRATUITY / TIP

NEEDED TAGS: 0x81, 0xDF63

EXAMPLE:
(EXT DEVICE -> CARD READER) >> 3E 3D 00 00 0F 01 01 81 04 00 00 00 1E DF 63 04 00 00 00 0A 25
(CARD READER -> EXT DEVICE) << 3E 00 00 00 00 3E

CLARIFICATION:
(EXT DEVICE -> CARD READER):
'>': 0x3E - Start packet.
CMD: 0x3D - Borica command.
00: 0x00 - Fixed value '00'.
LH LL: 0x00 0x0F - The data length is 15 bytes.
SUBCMD: 0x01 - TRANSACTION START subcommand.
[TRANS TYPE]: 0x01 - HEX format, PURCHASE

<TRANS PARAMS: VAR>: '81 04 00 00 00 1E DF 63 04 00 00 00 0A' - AMOUNT (tag 0x81) = 30 (0.30) and GRATUITY / TIP (tag 0xDF63) = 10 (0.10)

CSUM: 0x25 - Xor of all bytes (3E 3D 00 ..... 00 0A).

(CARD READER -> EXT DEVICE):
'>': 0x3E - Start packet.
00: 0x00 - Fixed value '00'.
ST: 0x00 - No error (ERROR CODES).
LH LL : 0x00 0x00 - The data length is 0 bytes.
CSUM: 0x3E - Xor of all bytes (3E 00 00 00 00).

Note: Tag 0x81 should contain the total <Amount> + <Gratuity>. If you need to start a transaction for total amount = 1.10 (1.00 amount + 0.10 Gratuity), then tag 0x81 should have value = '00 00 00 6E' and tag 0xDF63 should have value = ' 00 00 00 0A'.

3. 0x02: START PURCHASE + CASHBACK

NEEDED TAGS:

0x81, 0x9F04

EXAMPLE:

(EXT DEVICE -> CARD READER) >>	3E 3D 00 00 0F 01 02 81 04 00 00 00 32 9F 04 04 00 00 00 14 33
(CARD READER -> EXT DEVICE) <<	3E 00 00 00 00 3E

CLARIFICATION:

(EXT DEVICE -> CARD READER):

'>':	0x3E - Start packet.
CMD:	0x3D - BORICA CMD command.
00:	0x00 - Fixed value '00'.
LH LL:	0x00 0x0F - The data length is 15 bytes.
SUBCMD:	0x01 - TRANSACTION START subcommand.
[TRANS TYPE]:	0x02 - HEX format, PURCHASE + CASHBACK
<TRANS PARAMS: VAR>:	'81 04 00 00 00 32 9F 04 04 00 00 00 14' - AMOUNT (tag 0x81) = 50 (0.50) and CASHBACK (tag 9F04) = 20 (0.20)
CSUM:	0x33 - Xor of all bytes (3E 3D 00 ..... 00 14).

(CARD READER -> EXT DEVICE):

'>':	0x3E - Start packet.
00:	0x00 - Fixed value '00'.
ST:	0x00 - No error (ERROR CODES).
LH LL :	0x00 0x00 - The data length is 0 bytes.
CSUM:	0x3E - Xor of all bytes (3E 00 00 00 00).

Note: Tag 0x81 should contain only the <Amount>. If you need to start a transaction for total amount = 1.10 (1.00 amount + 0.10 Cashback), then tag 0x81 should have value = '00 00 00 64' and tag 0x9F04 should have value = '00 00 00 0A'.

4. 0x03: START PURCHASE + REFERENCE

NEEDED TAGS:

0x81, 0xDF03

EXAMPLE:

(EXT DEVICE -> CARD READER) >>	3E 3D 00 00 15 01 03 81 04 00 00 00 64 DF 03 0A 31 32 33 34 35 36 37 38 39 30 22
(CARD READER -> EXT DEVICE) <<	3E 00 00 00 00 3E

CLARIFICATION:

(EXT DEVICE -> CARD READER):

'>':	0x3E - Start packet.
CMD:	0x3D - BORICA CMD command.

00: 0x00 - Fixed value '00'.
LH LL: 0x00 0x15 - The data length is 21 bytes.
SUBCMD: 0x01 - TRANSACTION START subcommand.
[TRANS TYPE]: 0x03 - HEX format, PURCHASE + REFERENCE

<TRANS PARAMS: VAR>: '81 04 00 00 00 64 DF 03 0A 31 32 33 34 35 36 37 38 39 30' - AMOUNT (tag 0x81) = 100 (1.00) and REFERENCE (tag DF03) = "1234567890"

CSUM: 0x22 - Xor of all bytes (3E 3D 00 ..... 39 30).

(CARD READER -> EXT DEVICE):
'>': 0x3E - Start packet.
00: 0x00 - Fixed value '00'.
ST: 0x00 - No error (ERROR CODES).
LH LL : 0x00 0x00 - The data length is 0 bytes.
CSUM: 0x3E - Xor of all bytes (3E 00 00 00 00).

5. 0x04: START CASH ADVANCE

This transaction means cash withdrawals (such as ATM withdrawals).

NEEDED TAGS:
0x81

EXAMPLE:
(EXT DEVICE -> CARD READER) >> 3E 3D 00 00 08 01 04 81 04 00 00 00 64 EF
(CARD READER -> EXT DEVICE) << 3E 00 00 00 00 3E

CLARIFICATION:
(EXT DEVICE -> CARD READER):
'>': 0x3E - Start packet.
CMD: 0x3D - Borica command.
00: 0x00 - Fixed value '00'.
LH LL: 0x00 0x08 - The data length is 8 bytes.
SUBCMD: 0x01 - TRANSACTION START subcommand.
[TRANS TYPE]: 0x04 - HEX format, CASH ADVANCE

<TRANS PARAMS: VAR>: '81 04 00 00 00 64' - AMOUNT (tag 0x81) = 100 (1.00)

CSUM: 0xEF - Xor of all bytes (3E 3D 00 ..... 00 64).

(CARD READER -> EXT DEVICE):
'>': 0x3E - Start packet.
00: 0x00 - Fixed value '00'.
ST: 0x00 - No error (ERROR CODES).
LH LL : 0x00 0x00 - The data length is 0 bytes.
CSUM: 0x3E - Xor of all bytes (3E 00 00 00 00).

6. 0x05: START AUTHORIZATION

This transaction doesn't withdraw the money from the account. It only blocks them. After that the money can be returned with VOID OF AUTHORIZATION, or they can finally be withdrawed with the transaction PURCHASE + CODE. These types of transactions are used in Hotels for example.

NEEDED TAGS:
0x81

EXAMPLE:
(EXT DEVICE -> CARD READER) >> 3E 3D 00 00 08 01 05 81 04 00 00 00 64 EE
(CARD READER -> EXT DEVICE) << 3E 00 00 00 00 3E

CLARIFICATION:
(EXT DEVICE -> CARD READER):
'>': 0x3E - Start packet.
CMD: 0x3D - Borica command.
00: 0x00 - Fixed value '00'.
LH LL: 0x00 0x08 - The data length is 8 bytes.
SUBCMD: 0x01 - TRANSACTION START subcommand.
[TRANS TYPE]: 0x05 - HEX format, AUTHORIZATION

<TRANS PARAMS: VAR>: '81 04 00 00 00 64' - AMOUNT (tag 0x81) = 100 (1.00)

CSUM: 0xEE - Xor of all bytes (3E 3D 00 ..... 00 64).

(CARD READER -> EXT DEVICE):
'>': 0x3E - Start packet.
00: 0x00 - Fixed value '00'.
ST: 0x00 - No error (ERROR CODES).
LH LL : 0x00 0x00 - The data length is 0 bytes.
CSUM: 0x3E - Xor of all bytes (3E 00 00 00 00).

7. 0x06: START PURCHASE + CODE

When AUTHORIZATION has been done successfully, the Amount of the transaction was blocked, but still not withdrawed from the account. In order to withdraw the blocked Amount, PURCHASE + CODE needs to be done. That is why this type of transaction can successfully be done only after successful AUTHORIZATION before it. When the AUTHORIZATION is being approved, unique RRN and Authorization ID are returned from the host. They should be printed on the receipt. Then when starting PURCHASE + CODE, these RRN and Authorization ID from the first transaction need to be used in the command for starting a transaction. The amount also needs to be the same as in the AUTHORIZATION.

NEEDED TAGS: 0x81, 0xDF01, 0xDF02

EXAMPLE:

(EXT DEVICE -> CARD READER) >> 3E 3D 00 00 1B 01 06 81 04 00 00 00 64 DF 01 07 36 36 38 30 36 33 35 DF 02 06 39 32 30 32 31 32 CE

(CARD READER -> EXT DEVICE) << 3E 00 00 00 00 3E

CLARIFICATION:

(EXT DEVICE -> CARD READER):

'>':	0x3E - Start packet.
CMD:	0x3D - Borica command.
00:	0x00 - Fixed value '00'.
LH LL:	0x00 0x1B - The data length is 27 bytes.
SUBCMD:	0x01 - TRANSACTION START subcommand.
[TRANS TYPE]:	0x06 - HEX format, PURCHASE + CODE
<TRANS PARAMS: VAR>:	'81 04 00 00 00 64 DF 01 07 36 36 38 30 36 33 35 DF 02 06 39 32 30 32 31 32' - AMOUNT (tag 0x81) = 100 (1.00), RRN (tag 0xDF01) = "6680635", Authorization ID (tag 0xDF02) = "920212"
CSUM:	0xCE - Xor of all bytes (3E 3D 00 ..... 31 32).

(CARD READER -> EXT DEVICE):

'>':	0x3E - Start packet.
00:	0x00 - Fixed value '00'.
ST:	0x00 - No error (ERROR CODES).
LH LL :	0x00 0x00 - The data length is 0 bytes.
CSUM:	0x3E - Xor of all bytes (3E 00 00 00 00).

8. 0x07: START VOID OF PURCHASE

Voiding a Purchase. The Amount, RRN and Authorization ID, which are used here, need to be taken from the Purchase, which will be voided. After sending the command, if the PINPAD discovers a transaction, saved in the memory, which has the same Amount, RRN and Authorization ID as those, which are sent in the command, the CARD READER will send the Purchase for voiding without waiting for a card. But if the CARD READER doesn't discover such a transaction in the memory, it will prompt for a card and after this it will send the data for voiding to the server. This way it can void Purchase from previous days.

NEEDED TAGS:

0x81, 0xDF01, 0xDF02

EXAMPLE:

(EXT DEVICE -> CARD READER) >> 3E 3D 00 00 1B 01 07 81 04 00 00 00 64 DF 01 07 36 36 38 30 36 33 35 DF 02 06 39 32 30 32 31 32 CF

(CARD READER -> EXT DEVICE) << 3E 00 00 00 00 3E

CLARIFICATION:

(EXT DEVICE -> CARD READER):

'>':	0x3E - Start packet.
CMD:	0x3D - Borica command.
00:	0x00 - Fixed value '00'.
LH LL:	0x00 0x1B - The data length is 27 bytes.
SUBCMD:	0x01 - TRANSACTION START subcommand.
[TRANS TYPE]:	0x07 - HEX format, VOID OF PURCHASE

<TRANS PARAMS: VAR>: '81 04 00 00 00 64 DF 01 07 36 36 38 30 36 33 35 DF 02 06 39 32 30 32 31 32' - AMOUNT (tag 0x81) = 100 (1.00), RRN (tag 0xDF01) = "6680635", Authorization ID (tag 0xDF02) = "920212"

CSUM: 0xCF - Xor of all bytes (3E 3D 00 ..... 31 32).

(CARD READER -> EXT DEVICE):

'>':	0x3E - Start packet.
00:	0x00 - Fixed value '00'.
ST:	0x00 - No error (ERROR CODES).
LH LL :	0x00 0x00 - The data length is 0 bytes.
CSUM:	0x3E - Xor of all bytes (3E 00 00 00 00).

9. 0x07: START VOID OF PURCHASE + GRATUITY / TIP

Voiding a Purchase + Gratuity. The Amount, Gratuity, RRN and Authorization ID, which are used here, need to be taken from the Purchase + Gratuity, which will be voided. After sending the command, if the CARD READER discovers a transaction, saved in the memory, which has the same Amount, Gratuity, RRN and Authorization ID as those, which are sent in the command, the CARD READER will send the Purchase for voiding without waiting for a card. But if the CARD READER doesn't discover such a transaction in the memory, it will prompt for a card and after this it will send the data for voiding to the server. This way it can void Purchase from previous days.

NEEDED TAGS:

0x81, 0xDF63, 0xDF01, 0xDF02

EXAMPLE:

(EXT DEVICE -> CARD READER) >>	3E 3D 00 00 22 01 07 81 04 00 00 00 64 DF 63 04 00 00 00 0A DF 01 07 36 36 38 30 36 33 35 DF 02 06 39 32 30 32 31 32 44
(CARD READER -> EXT DEVICE) <<	3E 00 00 00 00 3E

CLARIFICATION:

(EXT DEVICE -> CARD READER):

'>':	0x3E - Start packet.
CMD:	0x3D - Borica command.
00:	0x00 - Fixed value '00'.
LH LL:	0x00 0x22 - The data length is 34 bytes.
SUBCMD:	0x01 - TRANSACTION START subcommand.

[TRANS TYPE]: 0x07 - HEX format, VOID OF PURCHASE

<TRANS PARAMS: VAR>: '81 04 00 00 00 64 DF 63 04 00 00 00 0A DF 01 07 36 36 38 30 36 33 35 DF 02 06 39 32 30 32 31 32' - AMOUNT (tag 0x81) = 100 (1.00), Gratuity (tag 0xDF63) = 10 (0.10), RRN (tag 0xDF01) = "6680635", Authorization ID (tag 0xDF02) = "920212"

CSUM: 0x44 - Xor of all bytes (3E 3D 00 ..... 31 32).

(CARD READER -> EXT DEVICE):

'>': 0x3E - Start packet.
00: 0x00 - Fixed value '00'.
ST: 0x00 - No error (ERROR CODES).
LH LL : 0x00 0x00 - The data length is 0 bytes.
CSUM: 0x3E - Xor of all bytes (3E 00 00 00 00).

10. 0x07: START VOID OF PURCHASE + CASHBACK

Voiding a Purchase + Cashback. The Amount, Cashback, RRN and Authorization ID, which are used here, need to be taken from the Purchase + Cashback, which will be voided. After sending the command, if the PINPAD discovers a transaction, saved in the memory, which has the same Amount, RRN and Authorization ID as those, which are sent in the command, the CARD READER will send the Purchase for voiding without waiting for a card. But if the CARD READER doesn't discover such a transaction in the memory, it will prompt for a card and after this it will send the data for voiding to the server. This way it can void Purchase from previous days.

NEEDED TAGS:

0x81, 0x9F04, 0xDF01, 0xDF02

EXAMPLE:

(EXT DEVICE -> CARD READER) >> 3E 3D 00 00 22 01 07 81 04 00 00 00 32 9F 04 04 00 00 00 14 DF 01 07 36 36 38 30 36 33 35 DF 02 06 39 32 30 32 31 32 2B

(CARD READER -> EXT DEVICE) << 3E 00 00 00 00 3E

CLARIFICATION:

(EXT DEVICE -> CARD READER):

'>': 0x3E - Start packet.
CMD: 0x3D - Borica command.
00: 0x00 - Fixed value '00'.
LH LL: 0x00 0x22 - The data length is 34 bytes.
SUBCMD: 0x01 - TRANSACTION START subcommand.
[TRANS TYPE]: 0x07 - HEX format, VOID OF PURCHASE

<TRANS PARAMS: VAR>: '81 04 00 00 00 32 9F 04 04 00 00 00 14 DF 01 07 36 36 38 30 36 33 35 DF 02 06 39 32 30 32 31 32' - AMOUNT (tag 0x81) = 50

(0.50), Cashback (tag 0x9F04) = 20 (0.20), RRN (tag 0xDF01) = "6680635", Authorization ID (tag 0xDF02) = "920212"

CSUM: 0x2B - Xor of all bytes (3E 3D 00 ..... 31 32).

(CARD READER -> EXT DEVICE):

'>': 0x3E - Start packet.
00: 0x00 - Fixed value '00'.
ST: 0x00 - No error (ERROR CODES).
LH LL : 0x00 0x00 - The data length is 0 bytes.
CSUM: 0x3E - Xor of all bytes (3E 00 00 00 00).

11. 0x08: START VOID OF CASH ADVANCE

Voiding a Cash Advance. The Amount, RRN and Authorization ID, which are used here, need to be taken from the Cash Advance, which will be voided. After sending the command, if the CARD READER discovers a transaction, saved in the memory, which has the same Amount, RRN and Authorization ID as those, which are sent in the command, the CARD READER will send the Cash Advance for voiding without waiting for a card.

NEEDED TAGS:

0x81, 0xDF01, 0xDF02

EXAMPLE:

(EXT DEVICE -> CARD READER) >> 3E 3D 00 00 1B 01 08 81 04 00 00 00 64 DF 01 07 36 36 38 30 36 33 35 DF 02 06 39 32 30 32 31 32 C0

(CARD READER -> EXT DEVICE) << 3E 00 00 00 00 3E

CLARIFICATION:

(EXT DEVICE -> CARD READER):

'>': 0x3E - Start packet.
CMD: 0x3D - Borica command.
00: 0x00 - Fixed value '00'.
LH LL: 0x00 0x1B - The data length is 27 bytes.
SUBCMD: 0x01 - TRANSACTION START subcommand.
[TRANS TYPE]: 0x08 - HEX format, VOID OF CASH ADVANCE

<TRANS PARAMS: VAR>: '81 04 00 00 00 64 DF 01 07 36 36 38 30 36 33 35 DF 02 06 39 32 30 32 31 32' - AMOUNT (tag 0x81) = 100 (1.00), RRN (tag 0xDF01) = "6680635", Authorization ID (tag 0xDF02) = "920212"

CSUM: 0xC0 - Xor of all bytes (3E 3D 00 ..... 31 32).

(CARD READER -> EXT DEVICE):

'>': 0x3E - Start packet.
00: 0x00 - Fixed value '00'.
ST: 0x00 - No error (ERROR CODES).

LH LL : 0x00 0x00 - The data length is 0 bytes.
CSUM: 0x3E - Xor of all bytes (3E 00 00 00 00).

12. 0x09: START VOID OF AUTHORIZATION

Voiding an Authorization. The Amount, RRN and Authorization ID, which are used here, need to be taken from the Authorization, which will be voided. After sending the command, if the CARD READER discovers a transaction, saved in the memory, which has the same Amount, RRN and Authorization ID as those, which are sent in the command, the CARD READER will send the Authorization for voiding without waiting for a card. But if the CARD READER doesn't discover such a transaction in the memory, it will prompt for a card and after this it will send the data for voiding to the server. This way it can void Authorization from previous days.

NEEDED TAGS:

0x81, 0xDF01, 0xDF02

EXAMPLE:

(EXT DEVICE -> CARD READER) >> 3E 3D 00 00 1B 01 09 81 04 00 00 00 64 DF 01 07 36 36 38 30 36 33 35 DF 02 06 39 32 30 32 31 32 C1

(CARD READER -> EXT DEVICE) << 3E 00 00 00 00 3E

CLARIFICATION:

(EXT DEVICE -> CARD READER):

'>': 0x3E - Start of the packet.
CMD: 0x3D - Borica command.
00: 0x00 - Fixed value '00'.
LH LL: 0x00 0x1B - The data length is 27 bytes.
SUBCMD: 0x01 - TRANSACTION START subcommand.
[TRANS TYPE]: 0x09 - HEX format, VOID OF AUTHORIZATION

<TRANS PARAMS: VAR>: '81 04 00 00 00 64 DF 01 07 36 36 38 30 36 33 35 DF 02 06 39 32 30 32 31 32' - AMOUNT (tag 0x81) = 100 (1.00), RRN (tag 0xDF01) = "6680635", Authorization ID (tag 0xDF02) = "920212"

CSUM: 0xC1 - Xor of all bytes (3E 3D 00 ..... 31 32).

(CARD READER -> EXT DEVICE):

'>': 0x3E - Start of the packet.
00: 0x00 - Fixed value '00'.
ST: 0x00 - No error (ERROR CODES).
LH LL : 0x00 0x00 - The data length is 0 bytes.
CSUM: 0x3E - Xor of all bytes (3E 00 00 00 00).

13. 0x0A: START END OF DAY

Note: Before starting End of day and before printing the Daily Report receipt, containing all the sums, the command “0x1A: GET PINPAD STATUS” should be sent to the Card reader and if the response contains “'C': HANG TRANSACTION 'C'”, then a Test connection should be executed first (0x0E: START TEST CONNECTION). After the Test connection, the receipt can be now constructed and printed and End of day can be executed.

Before performing End Of Day, Daily Report receipt needs to be printed. It needs to be printed BEFORE the transaction, because if End Of Day is successful, it will delete all transactions from the log, and it will not be possible information for the transactions to be taken anymore. In order to print the receipt, (04: GET REPORT TAGS) command needs to be called multiple times. The first call needs to be with parameter (01 - FIRST RECORD) and after that it needs to be called in loop with parameter (02 - NEXT RECORDS) until the ST field in the response from the CARD READER becomes (08). If ST field from the response has value (08), this means, that the last transaction in the log was reached. For each transaction the transaction type and some other information needs to be taken, in order to calculate the transaction sums and the card sums (for transactions, different from TEST CONNECTION and END OF DAY, which are successful (TAG_DF05_TRANSACTION_RESULT value = 0) and which are not reversed transactions (BOR_TAG_TO_TYPE value != 1)). After performing End Of Day, it is recommended to print on the end of the Daily Report if the File was sent successfully or not.

NEEDED TAGS:

none

EXAMPLE:

(EXT DEVICE -> CARD READER) >> 3E 3D 00 00 02 01 0A 0A
(CARD READER -> EXT DEVICE) << 3E 00 00 00 00 3E

CLARIFICATION:

(EXT DEVICE -> CARD READER):

'>':	0x3E - Start of the packet.
CMD:	0x3D - Borica command.
00:	0x00 - Fixed value '00'.
LH LL:	0x00 0x02 - The data length is 2 bytes.
SUBCMD:	0x01 - TRANSACTION START subcommand.
[TRANS TYPE]:	0x0A - HEX format, END OF DAY
<TRANS PARAMS: VAR>:	none
CSUM:	0x0A - Xor of all bytes (3E 3D 00 00 02 01 0A).

(CARD READER -> EXT DEVICE):

'>':	0x3E - Start of the packet.
00:	0x00 - Fixed value '00'.
ST:	0x00 - No error (ERROR CODES).
LH LL :	0x00 0x00 - The data length is 0 bytes.
CSUM:	0x3E - Xor of all bytes (3E 00 00 00 00).

14. 0x0B: START LOYALTY BALANCE

NEEDED TAGS:

0x81 - needs to have value 0!

EXAMPLE:

(EXT DEVICE -> CARD READER) >> 3E 3D 00 00 08 01 0B 81 04 00 00 00 00 84
(CARD READER -> EXT DEVICE) << 3E 00 00 00 00 3E

CLARIFICATION:

(EXT DEVICE -> CARD READER):

'>':	0x3E - Start of the packet.
CMD:	0x3D - Borica command.
00:	0x00 - Fixed value '00'.
LH LL:	0x00 0x08 - The data length is 8 bytes.
SUBCMD:	0x01 - TRANSACTION START subcommand.
[TRANS TYPE]:	0x0B - HEX format, LOYALTY BALANCE

<TRANS PARAMS: VAR>: '81 04 00 00 00 00' - AMOUNT (tag 0x81) = 0

CSUM: 0x84 - Xor of all bytes (3E 3D 00 ..... 00 00).

(CARD READER -> EXT DEVICE):

'>':	0x3E - Start of the packet.
00:	0x00 - Fixed value '00'.
ST:	0x00 - No error (ERROR CODES).
LH LL :	0x00 0x00 - The data length is 0 bytes.
CSUM:	0x3E - Xor of all bytes (3E 00 00 00 00).

15. 0x0C: START LOYALTY SPEND

NEEDED TAGS:

0x81

EXAMPLE:

(EXT DEVICE -> CARD READER) >> 3E 3D 00 00 08 01 0C 81 04 00 00 00 3C BF
(CARD READER -> EXT DEVICE) << 3E 00 00 00 00 3E

CLARIFICATION:

(EXT DEVICE -> CARD READER):

'>':	0x3E - Start of the packet.
CMD:	0x3D - Borica command.
00:	0x00 - Fixed value '00'.
LH LL:	0x00 0x08 - The data length is 8 bytes.
SUBCMD:	0x01 - TRANSACTION START subcommand.
[TRANS TYPE]:	0x0C - HEX format, LOYALTY SPEND

<TRANS PARAMS: VAR>: '81 04 00 00 00 3C' - AMOUNT (tag 0x81) = 60 POINTS

CSUM: 0xBF - Xor of all bytes (3E 3D 00 ..... 00 3C).

(CARD READER -> EXT DEVICE):

'>': 0x3E - Start of the packet.
00: 0x00 - Fixed value '00'.
ST: 0x00 - No error (ERROR CODES).
LH LL : 0x00 0x00 - The data length is 0 bytes.
CSUM: 0x3E - Xor of all bytes (3E 00 00 00 00).

16. 0x0D: START VOID OF LOYALTY SPEND

NEEDED TAGS:

0x81, 0xDF01, 0xDF02

EXAMPLE:

(EXT DEVICE -> CARD READER) >> 3E 3D 00 00 1B 01 0D 81 04 00 00 00 3C DF 01 07 36 39 38 30 35 36 38 DF 02 06 39 32 30 32 31 35 9E

(CARD READER -> EXT DEVICE) << 3E 00 00 00 00 3E

CLARIFICATION:

(EXT DEVICE -> CARD READER):

'>': 0x3E - Start of the packet.
CMD: 0x3D - Borica command.
00: 0x00 - Fixed value '00'.
LH LL: 0x00 0x1B - The data length is 27 bytes.
SUBCMD: 0x01 - TRANSACTION START subcommand.
[TRANS TYPE]: 0x0D - HEX format, VOID OF LOYALTY SPEND

<TRANS PARAMS: VAR>: '81 04 00 00 00 3C DF 01 07 36 39 38 30 35 36 38 DF 02 06 39 32 30 32 31 35' - AMOUNT (tag 0x81) = 60 POINTS - AMOUNT (tag 0x81) = 60 POINTS, RRN (tag 0xDF01) = "6980568", Authorization ID (tag 0xDF02) = "920215"

CSUM: 0x9E - Xor of all bytes (3E 3D 00 ..... 31 35).

(CARD READER -> EXT DEVICE):

'>': 0x3E - Start of the packet.
00: 0x00 - Fixed value '00'.
ST: 0x00 - No error (ERROR CODES).
LH LL : 0x00 0x00 - The data length is 0 bytes.
CSUM: 0x3E - Xor of all bytes (3E 00 00 00 00).

17. 0x0E: START TEST CONNECTION

NEEDED TAGS:

none

EXAMPLE:

(EXT DEVICE -> CARD READER) >> 3E 3D 00 00 02 01 0E 0E
(CARD READER -> EXT DEVICE) << 3E 00 00 00 00 3E

CLARIFICATION:

(EXT DEVICE -> CARD READER):

'>':	0x3E - Start of the packet.
CMD:	0x3D - Borica command.
00:	0x00 - Fixed value '00'.
LH LL:	0x00 0x02 - The data length is 2 bytes.
SUBCMD:	0x01 - TRANSACTION START subcommand.
[TRANS TYPE]:	0x0E - HEX format, TEST CONNECTION
<TRANS PARAMS: VAR>:	none
CSUM:	0x0E - Xor of all bytes (3E 3D 00 00 02 01 0E).

(CARD READER -> EXT DEVICE):

'>':	0x3E - Start of the packet.
00:	0x00 - Fixed value '00'.
ST:	0x00 - No error (ERROR CODES).
LH LL :	0x00 0x00 - The data length is 0 bytes.
CSUM:	0x3E - Xor of all bytes (3E 00 00 00 00).

18. 0x0F: START TMS UPDATE

NEEDED TAGS:

none

EXAMPLE:

(EXT DEVICE -> CARD READER) >> 3E 3D 00 00 02 01 0F 0F
(CARD READER -> EXT DEVICE) << 3E 00 00 00 00 3E

CLARIFICATION:

(EXT DEVICE -> CARD READER):

'>':	0x3E - Start of the packet.
CMD:	0x3D - BORICA CMD command.
00:	0x00 - Fixed value '00'.
LH LL:	0x00 0x02 - The data length is 2 bytes.
SUBCMD:	0x01 - TRANSACTION START subcommand.
[TRANS TYPE]:	0x0F - HEX format, TMS UPDATE
<TRANS PARAMS: VAR>:	none
CSUM:	0x0F - Xor of all bytes (3E 3D 00 00 02 01 0F).

(CARD READER -> EXT DEVICE):

'>':	0x3E - Start of the packet.
00:	0x00 - Fixed value '00'.
ST:	0x00 - No error (ERROR CODES).
LH LL :	0x00 0x00 - The data length is 0 bytes.
CSUM:	0x3E - Xor of all bytes (3E 00 00 00 00).

3. 0x02: GET RECEIPT TAGS

DESCRIPTION:

At the end of the transaction, this command can be used to get information for the lastly performed transaction. This way a receipt can be printed.

EXT DEVICE:

3D 00 LH LL 02 <TAGS: VAR>

<TAGS: VAR>:

Tags to be returned. (For reference see "TLV STRUCTURE" and "TAGS" sections below).

CARD READER:

00 ST LH LL <DATA: VAR>

<DATA: VAR>:

Returned tags data

EXAMPLE:

If Purchase was performed and we want to get the amount and the transaction type, then we need to send the following command:

(EXT DEVICE -> CARD READER) >>	3E 3D 00 00 04 02 81 DF 10 4B
(CARD READER -> EXT DEVICE) <<	3E 00 08 00 00 36

CLARIFICATION:

(EXT DEVICE -> CARD READER):

'>':	0x3E - Start of the packet.
CMD:	0x3D - BORICA CMD command.
00:	0x00 - Fixed value '00'.
LH LL:	0x00 0x04 - The data length is 4 bytes.
SUBCMD:	0x02 - GET RECEIPT TAGS subcommand.
<TAGS: VAR>:	'81 DF 10' - Here we need to list all tags, which we would like to get. In this case we need to get 2 tags - 0x81 and 0xDF10.
CSUM:	0x4B - Xor of all bytes (3E 3D 00 ..... DF 10).

(CARD READER -> EXT DEVICE)

'>':	0x3E - Start of the packet.
00:	0x00 - Fixed value '00'.

ST:	0x08 - This command needs to be called only in the end of a transaction. In other cases it will return ST with value (08) - There is no data. In this case no data will be returned.
LH LL:	0x00 0x00 - The data length is 0 bytes.
<DATA: VAR>:	none
CSUM:	0x36 - Xor of all bytes (3E 00 08 00 00).

4. 0x03: TRANSACTION END

DESCRIPTION:

This command is being used if the EXT DEVICE needs externally to force quit the transaction, after it was started, or in the end of every transaction.

EXT DEVICE:

3D 00 LH LL 03 <CFM: 2>

<CFM: 2>:

This parameter is optional. If it is not present, then the "END OK" is used by default. If it is present, then it should have any of the following values:

[CFM HIGH][CFM LOW]:
00 00 - END FAIL
00 01 - END OK

CARD READER:

00 ST 00 00

EXAMPLE:

(EXT DEVICE -> CARD READER) >>	3E 3D 00 00 03 03 00 00 03
(CARD READER -> EXT DEVICE) <<	3E 00 00 00 00 3E

CLARIFICATION:

(EXT DEVICE -> CARD READER):

'>':	0x3E - Start of the packet.
CMD:	0x3D - Borica command.
00:	0x00 - Fixed value '00'.
LH LL:	0x00 0x03 - The data length is 3 bytes.
SUBCMD:	0x03 - TRANSACTION END subcommand.
<CFM: 2>	0x00 0x00 - END FAIL
CSUM:	0x03 - Xor of all bytes (3E 3D 00 00 03 03 00 00).

(CARD READER -> EXT DEVICE)

'>':	0x3E - Start of the packet.
00:	0x00 - Fixed value '00'.
ST:	0x00 - No error (ERROR CODES).
LH LL :	0x00 0x00 - The data length is 0 bytes.
CSUM:	0x3E - Xor of all bytes (3E 00 00 00 00).

5. 0x04: GET REPORT TAGS

This command is being used when we need to get some information for some specific transaction. We can get information for the last transaction (with parameter (03 - LAST RECORD)), then we can get information for all transactions one by one before the last (if we call the same command multiple times, but with parameter (04 - PREVIOUS RECORD). We can also get information for the first one (01 - FIRST RECORD) and then call the same command multiple times to get information for all next commands one by one (with parameter (02 - NEXT RECORDS)). It is similar to the command (02: GET RECEIPT TAGS).

EXT DEVICE:

3D 00 LH LL 04 <TYPE: 2> <TAGS: VAR>

<TYPE>

[TYPE HIGH][TYPE LOW]:
00 01 - FIRST RECORD
00 02 - NEXT RECORDS
00 03 - LAST RECORD
00 04 - PREVIOUS RECORD
00 10 - GET ALL RECORDS AS EVENTS

<TAGS>

Tags to be returned. (For reference see "TLV STRUCTURE" and "TAGS" sections below)

CARD READER:

00 ST LH LL <DATA: VAR>

EXAMPLE:

If we want to get the amount for the first transaction, then we need to send this command:

(EXT DEVICE -> CARD READER) >> 3E 3D 00 00 04 04 00 01 81 83
(CARD READER -> EXT DEVICE) << 3E 00 00 00 06 81 04 00 00 00 64 D9

CLARIFICATION:

(EXT DEVICE -> CARD READER):

'>':	0x3E - Start of the packet.
CMD:	0x3D - Borica command.
00:	0x00 - Fixed value '00'.
LH LL:	0x00 0x04 - The data length is 4 bytes.
SUBCMD:	0x04 - GET REPORT TAGS subcommand.
<TYPE: 2>:	0x00 0x01 - We want to get information for the FIRST RECORD.
<TAGS: VAR>:	0x81 - Here we need to list all tags, which we would like to get. In this case we need to get 1 tag - 0x81.
CSUM:	0x83 - Xor of all bytes (3E 3D 00 ..... 01 81).

(CARD READER -> EXT DEVICE):

'>':	0x3E - Start of the packet.
00:	0x00 - Fixed value '00'.
ST:	0x00 - No error (ERROR CODES).
LH LL :	0x00 0x06 - The data length is 6 bytes.

<DATA: VAR>:	'81 04 00 00 00 64' - The first transaction in this device has amount = 100 (1.00).
CSUM:	0xD9 - Xor of all bytes (3E 00 00 ..... 00 64).

6. 0x05: GET REPORT INFO

EXT DEVICE:

3D 00 00 01 05

CARD READER:

00 ST 00 02 <RECCNT: 2>

<RECCNT>

[RECCNT HIGH][RECCNT LOW]: THE COUNT OF RECORDS (0..65535)

EXAMPLE:

(EXT DEVICE -> CARD READER) >>	3E 3D 00 00 01 05 07
(CARD READER -> EXT DEVICE) <<	3E 00 00 00 02 01 6B 56

CLARIFICATION:

(EXT DEVICE -> CARD READER):

'>':	0x3E - Start of the packet.
CMD:	0x3D - Borica command.
00:	0x00 - Fixed value '00'.
LH LL:	0x00 0x01 - The data length is 1 byte.
SUBCMD:	0x05 - GET REPORT INFO subcommand.
CSUM:	0x07 - Xor of all bytes (3E 3D 00 00 01 05).

(CARD READER -> EXT DEVICE):

'>':	0x3E - Start of the packet.
00:	0x00 - Fixed value '00'.
ST:	0x00 - No error (ERROR CODES).
LH LL :	0x00 0x02 - The data length is 2 bytes.
<RECCNT: 2>:	0x01 0x6B - The count of records is 363.
CSUM:	0x56 - Xor of all bytes (3E 00 00 00 02 01 6B).

7. 0x06: GET PINPAD INFO

EXT DEVICE:

3D 00 00 01 06

CARD READER:

00 ST 00 2B <DATA: 43 BYTES>

<DATA>:

<MODEL NAME: 20 BYTES>:	MODEL OF THE PINPAD
<SERIAL NUMBER: 10 BYTES>:	SERIAL NUMBER OF THE PINPAD
<SOFT VER: 4 BYTES>:	SOFTWARE VERSION OF THE PINPAD
<TERMINAL ID: 8 BYTES>:	TERMINAL ID OF THE PINPAD
[MENU TYPE]:	TYPE OF THE MENU OF THE TERMINAL

BOR_MENU_TYPE_TRADING	1
BOR_MENU_TYPE_TRADING_REF	2
BOR_MENU_TYPE_TRADING_CASHBACK	3
BOR_MENU_TYPE_HOTEL_RENT_A_CAR	4
BOR_MENU_TYPE_BAR_RESTAURANT	5
BOR_MENU_TYPE_BANK_CHANGE	6
BOR_MENU_TYPE_TEST	7

EXAMPLE:

(EXT DEVICE -> CARD READER) >>	3E 3D 00 00 01 06 04
(CARD READER -> EXT DEVICE) <<	3E 00 00 00 2B 42 6C 75 65 50 61 64 2D 35 30 00 00 00 00 00 00 00 00 00 00 30 39 30 31 39 30 30 30 30 31 01 01 17 00 39 33 38 30 30 31 33 33 07 45

CLARIFICATION:

(EXT DEVICE -> CARD READER):

'>':	0x3E - Start of the packet.
CMD:	0x3D – Borica command.
00:	0x00 - Fixed value '00'.
LH LL:	0x00 0x01 - The data length is 1 byte.
SUBCMD:	0x06 - GET PINPAD INFO subcommand.
CSUM:	0x04 - Xor of all bytes (3E 3D 00 00 01 06).

(CARD READER -> EXT DEVICE)

'>':	0x3E - Start of the packet.
00:	0x00 - Fixed value '00'.
ST:	0x00 - No error (ERROR CODES).
LH LL :	0x00 0x2B - The data length is 43 bytes.
<MODEL NAME: 20 BYTES>:	'42 6C 75 65 50 61 64 2D 35 30 00 00 00 00 00 00 00 00 00 00' - STRING format, "BluePad-50"
<SERIAL NUMBER: 10 BYTES>:	'30 39 30 31 39 30 30 30 30 31' - STRING format, "0901900001"
<SOFT VER: 4 BYTES>:	'01 01 17 00' - HEX format, version 1.1.23.0
<TERMINAL ID: 8 BYTES>:	'39 33 38 30 30 31 33 33' - STRING format, "93800133"
[MENU TYPE]:	0x07 - HEX format, <BOR_MENU_TYPE_TEST>
CSUM:	0x45 - Xor of all bytes (3E 00 00 ..... 33 07).

8. 0x07: GET RTC

EXT DEVICE:

3D 00 00 01 07

CARD READER:

00 ST 00 06 year[1] month[1] date[1] hour[1] min[1] sec[1]

EXAMPLE:

(EXT DEVICE -> CARD READER) >> 3E 3D 00 00 01 07 05
(CARD READER -> EXT DEVICE) << 3E 00 00 00 06 13 03 14 09 30 2C 29

CLARIFICATION:

(EXT DEVICE -> CARD READER):

'>':	0x3E - Start of the packet.
CMD:	0x3D – Borica command.
00:	0x00 - Fixed value '00'.
LH LL:	0x00 0x01 - The data length is 1 byte.
SUBCMD:	0x07 - GET RTC subcommand.
CSUM:	0x05 - Xor of all bytes (3E 3D 00 00 01 07).

(CARD READER -> EXT DEVICE)

'>':	0x3E - Start of the packet.
00:	0x00 - Fixed value '00'.
ST:	0x00 - No error (ERROR CODES).
LH LL :	0x00 0x06 - The data length is 6 bytes.
year[1]:	0x13 - HEX format, year 19 (2019)
month[1]:	0x03 - HEX format, month 03
date[1]:	0x14 - HEX format, day 20
hour[1]:	0x09 - HEX format, hour 09
min[1]:	0x30 - HEX format, minute 48
sec[1]:	0x2C - HEX format, second 44
CSUM:	0x29 - Xor of all bytes (3E 00 00 ..... 30 2C).

9. 0x08: SET RTC

EXT DEVICE:

3D 00 00 07 08 year[1] month[1] date[1] hour[1] min[1] sec[1]

CARD READER:

00 ST 00 00

EXAMPLE:

(EXT DEVICE -> CARD READER) >> 3E 3D 00 00 07 08 13 03 14 09 30 2C 1D
(CARD READER -> EXT DEVICE) << 3E 00 00 00 00 3E

CLARIFICATION:

(EXT DEVICE -> CARD READER)

'>':	0x3E - Start of the packet.
CMD:	0x3D - Borica command.
00:	0x00 - Fixed value '00'.
LH LL:	0x00 0x07 - The data length is 7 bytes.
SUBCMD:	0x08 - SET RTC subcommand.
year[1]:	0x13 - HEX format, year 19 (2019)
month[1]:	0x03 - HEX format, month 03
date[1]:	0x14 - HEX format, day 20
hour[1]:	0x09 - HEX format, hour 09
min[1]:	0x30 - HEX format, minute 48
sec[1]:	0x2C - HEX format, second 44
CSUM:	0x1D - Xor of all bytes (3E 3D 00 ..... 30 2C).

(CARD READER -> EXT DEVICE)

'>':	0x3E - Start of the packet.
00:	0x00 - Fixed value '00'.
ST:	0x00 - No error (ERROR CODES).
LH LL :	0x00 0x00 - The data length is 0 bytes.
CSUM:	0x3E - Xor of all bytes (3E 00 00 00 00).

10. 0x0B: DELETE BATCH

DESCRIPTION:

Use this command to delete the batch. This is emergency operation! In real conditions it should not be performed, except in some rare and emergency cases! This is why this operation is protected with password, which needs to be manually entered from the merchant.

EXT DEVICE:

3D 00 LH LL 0B <PASSWORD: VAR>

CARD READER:

00 ST 00 00

EXAMPLE:

(EXT DEVICE -> CARD READER) >>	3E 3D 00 00 0A 0B 30 31 32 33 34 33 32 31 30 36
(CARD READER -> EXT DEVICE) <<	3E 00 00 00 00 3E

CLARIFICATION:

(EXT DEVICE -> CARD READER)

'>':	0x3E - Start of the packet.
CMD:	0x3D - Borica command.
00:	0x00 - Fixed value '00'.
LH LL:	0x00 0x0A - The data length is 10 bytes.
SUBCMD:	0x0B - DELETE BATCH subcommand.
CSUM:	0x36 - Xor of all bytes (3E 3D 00 00 ... 32 31 30).

(CARD READER -> EXT DEVICE)

'>':	0x3E - Start of the packet.
00:	0x00 - Fixed value '00'.
ST:	0x00 - No error (ERROR CODES).
LH LL :	0x00 0x00 - The is no data.
CSUM:	0x3E - Xor of all bytes (3E 00 00 00 00).

11. 0x0C: CLEAR REVERSAL

DESCRIPTION:

Use this command to delete a reversal. This is emergency operation! In real conditions it should not be performed, except in some rare and emergency cases! This is why this operation is protected with password, which needs to be manually entered from the merchant.

EXT DEVICE:

3D 00 LH LL 0C <PASSWORD: VAR>

CARD READER:

00 ST 00 00

EXAMPLE:

(EXT DEVICE -> CARD READER) >>	3E 3D 00 00 0A 0C 30 31 32 33 34 33 32 31 30 31
(CARD READER -> EXT DEVICE) <<	3E 00 00 00 00 3E

CLARIFICATION:

(EXT DEVICE -> CARD READER)

'>':	0x3E - Start of the packet.
CMD:	0x3D - Borica command.
00:	0x00 - Fixed value '00'.
LH LL:	0x00 0x0A - The data length is 10 bytes.
SUBCMD:	0x0C - CLEAR REVERSAL subcommand.
CSUM:	0x31 - Xor of all bytes (3E 3D 00 00 ... 32 31 30).

(CARD READER -> EXT DEVICE)

'>':	0x3E - Start of the packet.
00:	0x00 - Fixed value '00'.
ST:	0x00 - No error (ERROR CODES).
LH LL :	0x00 0x00 - The is no data.
CSUM:	0x3E - Xor of all bytes (3E 00 00 00 00).

12. 0x1A: GET PINPAD STATUS

EXT DEVICE:

3D 00 00 01 1A

CARD READER:

00 ST 00 02 <REVERSAL> <END DAY>

<REVERSAL>: Shows if there is any transaction, which needs to be reversed:

00: NO REVERSAL
'R': REVERSAL 'R'
'C': HANG TRANSACTION 'C'

<END DAY>: Shows if end of day needs to be performed before any other type of transaction:

00: NO END OF DAY REQUIRED
01: YOU NEED TO PERFORM END OF DAY

EXAMPLE:

(EXT DEVICE -> CARD READER) >> 3E 3D 00 00 01 1A 18
(CARD READER -> EXT DEVICE) << 3E 00 00 00 02 00 00 3C

CLARIFICATION:

(EXT DEVICE -> CARD READER)

'>':	0x3E - Start of the packet.
CMD:	0x3D - Borica command.
00:	0x00 - Fixed value '00'.
LH LL:	0x00 0x01 - The data length is 1 byte.
SUBCMD:	0x1A - GET PINPAD STATUS subcommand.
CSUM:	0x18 - Xor of all bytes (3E 3D 00 00 01 1A).

(CARD READER -> EXT DEVICE)

'>':	0x3E - Start of the packet.
00:	0x00 - Fixed value '00'.
ST:	0x00 - No error (ERROR CODES).
LH LL :	0x00 0x02 - The data length is 2 bytes.
<REVERSAL>:	0x00 - HEX format, No transaction needs to be reversed
<END DAY>:	0x00 - HEX format, No end of day required
CSUM:	0x3C - Xor of all bytes (3E 00 00 00 02 00 00).

13. 0x1D: GET MENU INFO

EXT DEVICE:

3D 00 00 01 1D

CARD READER:

00 ST 00 7A <DATA: 122 BYTES>

<DATA>:

<SERIAL NUMBER: 10 BYTES>:	SERIAL NUMBER OF THE PINPAD
<TERMINAL ID: 8 BYTES>:	TERMINAL ID OF THE PINPAD
<MERCHANT ID: 16 BYTES>:	MERCHANT ID OF THE PINPAD

<SOFT VER: 4 BYTES>:	SOFTWARE VERSION OF THE PINPAD
<COMP DATE: 16 BYTES>:	COMPILATION DATE OF THE APPLICATION
<COMP TIME: 10 BYTES>:	COMPILATION TIME OF THE APPLICATION
<AUTH IP: 16 BYTES>:	AUTHORIZATION IP
<AUTH PORT: 2 BYTES>:	AUTHORIZATION PORT
<KEY LOAD IP: 16 BYTES>:	KEY LOAD IP
<KEY LOAD PORT: 2 BYTES>:	KEY LOAD PORT
<TMS IP: 16 BYTES>:	TMS IP
<TMS PORT: 2 BYTES>:	TMS PORT
<AUTH NII HOST: 2 BYTES>:	AUTH NII HOST
<KMC NII HOST: 2 BYTES>:	KMC NII HOST
<KEYS STATE: 1 BYTE>:	SYSTEM KEYS STATE

0x00: “NO KEYS”
0x01: “SYSTEM KEK ONLY”
0x02: “KEK & MAC OK”
0x03: “MAN KEK ONLY”
0x04: “UNKNOWN KEYS”
0x05: “INTERNAL ERROR”

EXAMPLE:

(EXT DEVICE -> CARD READER) >>	3E 3D 00 00 01 1D 1F
(CARD READER -> EXT DEVICE) <<	3E 00 00 00 7B 30 39 32 30 39 30 30 30 30 34 39 33 38 30 30 34 31 38 39 39 39 33 39 30 30 30 30 31 30 30 30 30 30 00 01 01 1C 00 4A 61 6E 20 32 37 20 32 30 32 31 00 00 00 00 00 31 31 3A 35 34 3A 35 30 00 00 31 39 33 2E 34 31 2E 31 39 30 2E 31 37 30 00 00 4E EE 31 39 33 2E 34 31 2E 31 39 30 2E 31 37 30 00 00 4E EE 31 39 33 2E 34 31 2E 31 39 30 2E 31 36 39 00 00 1F 40 03 33 06 21 02 7B

CLARIFICATION:

(EXT DEVICE -> CARD READER):

'>':	0x3E - Start of the packet.
CMD:	0x3D – Borica command.
00:	0x00 - Fixed value '00'.
LH LL:	0x00 0x01 - The data length is 1 byte.
SUBCMD:	0x1D - GET MENU INFO subcommand.
CSUM:	0x1F - Xor of all bytes (3E 3D 00 00 01 1D).

(CARD READER -> EXT DEVICE)

'>':	0x3E - Start of the packet.
00:	0x00 - Fixed value '00'.
ST:	0x00 - No error (ERROR CODES).
LH LL :	0x00 0x7B - The data length is 123 bytes.

<SERIAL NUMBER: 10 BYTES>:	' 30 39 32 30 39 30 30 30 30 34' – STRING format, “0920900004”
<TERMINAL ID: 8 BYTES>:	‘39 33 38 30 30 34 31 38’ - STRING format, “93800418”
<MERCHANT ID: 16 BYTES>:	‘39 39 39 33 39 30 30 30 30 31 30 30 30 30 30 00’ - STRING format, “999390000100000”
<SOFT VER: 4 BYTES>:	'01 01 1C 00' – HEX format, version 1.1.28.0
<COMP DATE: 16 BYTES>:	'4A 61 6E 20 32 37 20 32 30 32 31 00 00 00 00 00' – STRING format, “Jan 27 2021”
<COMP TIME: 10 BYTES>:	‘31 31 3A 35 34 3A 35 30 00 00’ - STRING format, “11:54:50”
<AUTH IP: 16 BYTES>:	‘31 39 33 2E 34 31 2E 31 39 30 2E 31 37 30 00 00’ - STRING format, “193.41.190.170”
<AUTH PORT: 2 BYTES>:	‘4E EE’ - HEX format, Port = 20206
<KEY LOAD IP: 16 BYTES>:	‘31 39 33 2E 34 31 2E 31 39 30 2E 31 37 30 00 00’ - STRING format, “193.41.190.170”
<KEY LOAD PORT: 2 BYTES>:	‘4E EE’ - HEX format, Port = 20206
<TMS IP: 16 BYTES>:	‘31 39 33 2E 34 31 2E 31 39 30 2E 31 36 39 00 00’ - STRING format, “193.41.190.169”.
<TMS PORT: 2 BYTES>:	‘1F 40’ - HEX format, Port = 8000
<AUTH NII HOST: 2 BYTES>:	‘03 33’ - BCD format, ANII = 333
<KMC NII HOST: 2 BYTES>:	‘06 21’ - BCD format, KNII = 621
<KEYS STATE: 1 BYTE>:	0x02 - “KEK & MAC OK”
CSUM:	0x7B - Xor of all bytes (3E 00 00 00 … 33 06 21).

14. 0x1E: GET PUBLIC KEYS LIST

EXT DEVICE:

3D 00 00 01 1E

CARD READER:

00 ST LH LL <DATA: VAR>

<DATA: VAR>: Data in string format. Just print the string.

EXAMPLE:

(EXT DEVICE -> CARD READER) >>	3E 3D 00 00 01 1E 1C
(CARD READER -> EXT DEVICE) <<	3E 00 00 02 52 20 31 20 41 30 30 30 30 30 30 30 30 33 20 30 37 20 31 32 32 36 0A 20 32 20 41 30 30 30 30 30 30 30 30 33 20 30 38 20 31 32 32 34 0A 20 33 20 41 30 30 30 30 30 30 30 30 33 20 30 39 20 31 32 32 36 0A 20 34 20 41 30 30 30 30 30
	..... (too long response) .....
	32 37 20 41 30 30 30 30 30 30 34 36 31 20 41 32 20 31 32 32 34 0A 2A

CLARIFICATION:

(EXT DEVICE -> CARD READER):

'>':	0x3E - Start of the packet.
CMD:	0x3D – Borica command.
00:	0x00 - Fixed value '00'.
LH LL:	0x00 0x01 - The data length is 1 byte.
SUBCMD:	0x1E - GET PUBLIC KEYS LIST subcommand.
CSUM:	0x1C - Xor of all bytes (3E 3D 00 00 01 1E).

(CARD READER -> EXT DEVICE)

'>':	0x3E - Start of the packet.
00:	0x00 - Fixed value '00'.
ST:	0x00 - No error (ERROR CODES).
LH LL :	0x02 0x52 - The data length is 594 bytes.
<DATA: VAR>:	'20 31 20 41 30 … 31 32 32 34 0A' - Just print the string
CSUM:	0x2A - Xor of all bytes (3E 00 00 00 … 33 06 21).

15. 0x1F: GET SYMMETRIC KEYS LIST

EXT DEVICE:

3D 00 00 01 1F

CARD READER:

00 ST LH LL <DATA: VAR>

<DATA: VAR>: Data in string format. Just print the string.

EXAMPLE:

(EXT DEVICE -> CARD READER) >>	3E 3D 00 00 01 1F 1D
(CARD READER -> EXT DEVICE) <<	3E 00 00 00 64 53 59 53 54 45 4D 20 4B 45 4B 3A 20 20 35 37 44 35 45 37 0A 53 59 53 54 45 4D 20 4D 41 43 3A 20 20 45 33 44 42 42 43 0A 4D 41 53 54 45 52 20 50 49 4E 3A 20 20 43 43 31 35 46 34 0A 4D 41 53 54 45 52 20 4D 41 43 3A 20 20 38 33 44 36 44 39 0A 4D 41 53 54 45 52 20 44 41 54 41 3A 20 35 35 42 34 33 46 0A 75

CLARIFICATION:

(EXT DEVICE -> CARD READER):

'>':	0x3E - Start of the packet.
CMD:	0x3D – Borica command.
00:	0x00 - Fixed value '00'.
LH LL:	0x00 0x01 - The data length is 1 byte.
SUBCMD:	0x1F - GET SYMMETRIC KEYS LIST subcommand.
CSUM:	0x1D - Xor of all bytes (3E 3D 00 00 01 1F).

(CARD READER -> EXT DEVICE)

'>':	0x3E - Start of the packet.
00:	0x00 - Fixed value '00'.
ST:	0x00 - No error (ERROR CODES).

LH LL :	0x00 0x64 - The data length is 100 bytes.
<DATA: VAR>:	‘53 59 53 54 45 … 42 34 33 46 0A’ - Just print the string
CSUM:	0x75 - Xor of all bytes (3E 00 00 00 … 33 46 0A).

16. 0x20: EDIT COMMUNICATION PARAMETERS

DESCRIPTION:

Use this command to manually update one of the communication parameters, listed below. The password has to be manually entered from the user! It is not hardcoded parameter of the command.

EXT DEVICE:

3D 00 LH LL 20 <DATA: VAR>

<DATA: VAR>: Data in TLV format.

DF8003 :	(Mandatory tag!) This is the password, which has to be manually entered from the user.
DF51 :	(Optional) Host IP
DF52 :	(Optional) Host port
DF7B :	(Optional) Key load IP
DF7C :	(Optional) Key load port
DF65 :	(Optional) TMS IP
DF66 :	(Optional) TMS port

CARD READER:

00 ST 00 00

EXAMPLE:

(EXT DEVICE -> CARD READER) >>	3E 3D 00 00 3A 20 DF 80 03 09 30 31 32 33 34 33 32 31 30 DF 51 0E 31 39 33 2E 34 31 2E 31 39 30 2E 31 37 30 DF 52 02 4E EE DF 7B 0E 31 39 33 2E 34 31 2E 31 39 30 2E 31 37 30 DF 7C 02 4E EE 7C
(CARD READER -> EXT DEVICE) <<	3E 00 00 00 00 3E

CLARIFICATION:

(EXT DEVICE -> CARD READER):

'>':	0x3E - Start of the packet.
CMD:	0x3D – Borica command.
00:	0x00 - Fixed value '00'.
LH LL:	0x00 0x3A - The data length is 58 bytes.
SUBCMD:	0x20 - EDIT COMM PARAMETERS subcommand.
<DATA: VAR>:	‘DF 80 03 … 02 4E EE’ - Data in TLV format.
CSUM:	0x7C - Xor of all bytes (3E 3D 00 00 01 1F).

(CARD READER -> EXT DEVICE)

'>':	0x3E - Start of the packet.
00:	0x00 - Fixed value '00'.
ST:	0x00 - No error (ERROR CODES).

LH LL :	0x00 0x00 - The data length is 0 bytes.
CSUM:	0x3E - Xor of all bytes (3E 00 00 00 00).

17. 0x21: KEYS COMMAND

DESCRIPTION:

This command performs few operations with the keys – either Init, Uptade or Decommission. These 3 operations can be performed either with HSM, connected via USB cable or with remote connection to the KMC.

After this command is executed successfully, the card reader will start the requested operation. In case of 0x01 (Init keys), 0x03 (Update keys) and 0x05 (Decommission keys), the card reader will send data directly to the HSM, so it is expected that there will be an USB connection and this data will just be redirected through the USB. In case of 0x02 (Init keys remote), 0x04 (Update keys remote) and 0x06 (Decommission keys remote), the card reader will send the data to the remote KMC. The communication in this case will be very similar to a normal transaction – first the card reader will send event for opening a socket to the host, then it will send an event for sending data to the host and etc.

EXT DEVICE:

3D 00 00 02 21 <OPERATION TYPE: 1 BYTE>

<OPERATION TYPE: 1 BYTE> Specify the operation type

0x01:	INIT KEYS
0x02:	INIT KEYS REMOTE
0x03:	UPDATE KEYS
0x04:	UPDATE KEYS REMOTE
0x05:	DECOMMISSION KEYS
0x06:	DECOMMISSION KEYS REMOTE

CARD READER:

00 ST 00 00

EXAMPLE:

(EXT DEVICE -> CARD READER) >>	3E 3D 00 00 02 21 01 21
(CARD READER -> EXT DEVICE) <<	3E 00 00 00 00 3E

CLARIFICATION:

(EXT DEVICE -> CARD READER):

'>':	0x3E - Start of the packet.
CMD:	0x3D – Borica command.
00:	0x00 - Fixed value '00'.
LH LL:	0x00 0x02 - The data length is 2 bytes.
SUBCMD:	0x20 - KEYS COMMAND subcommand.
<DATA: VAR>:	0x01 – Start INIT KEYS operation
CSUM:	0x21 - Xor of all bytes (3E 3D 00 00 02 21 01).

(CARD READER -> EXT DEVICE)

'>':	0x3E - Start of the packet.
00:	0x00 - Fixed value '00'.
ST:	0x00 - No error (ERROR CODES).
LH LL :	0x00 0x00 - The data length is 0 bytes.
CSUM:	0x3E - Xor of all bytes (3E 00 00 00 00).

18. 0x23: CHECK PASSWORD

DESCRIPTION:

Check if the password is correct.

EXT DEVICE:

3D 00 LH LL 23 [PASSWORD TYPE: 1 BYTE] <PASSWORD: VAR>

[PASSWORD TYPE: 1 BYTE]

0x01 – Merchant password
0x02 – Bank passwords
0x03 – Custom password 1
0x04 – Custom password 2
0x05 – Custom password 3
0x06 – Internal password

CARD READER:

00 ST 00 00

EXAMPLE:

(EXT DEVICE -> CARD READER) >>	3E 3D 00 00 06 23 01 31 30 30 30 26
(CARD READER -> EXT DEVICE) <<	3E 00 15 00 00 2B

CLARIFICATION:

(EXT DEVICE -> CARD READER):

'>':	0x3E - Start of the packet.
CMD:	0x3D – Borica command.
00:	0x00 - Fixed value '00'.
LH LL:	0x00 0x06 - The data length is 6 bytes.
SUBCMD:	0x23 – CHECK PASSWORD subcommand.
[PASSWORD TYPE]	0x01 – Check Merchant password.
<PASSWORD: VAR>	‘31 30 30 30’ - String format, “1000”.
CSUM:	0x26 - Xor of all bytes (3E 3D 00 … 30 30 30).

(CARD READER -> EXT DEVICE)

'>':	0x3E - Start of the packet.
00:	0x00 - Fixed value '00'.
ST:	0x15 - errInvPass (ERROR CODES) – Wrong password.
LH LL :	0x00 0x00 - The data length is 0 bytes.
CSUM:	0x2B - Xor of all bytes (3E 00 15 00 00).

19. 0x24: GET REPORT TAGS BY STAN

This command is being used in case that it is needed a information for some specific transaction to be taken. To specify the transaction, its STAN (transaction number) needs to be used.

EXT DEVICE:

3D 00 LH LL 24 <STAN: 4> <TAGS: VAR>

<STAN> The STAN of the record as a big endian integer.

<TAGS> Tags to be returned (For reference see "TLV STRUCTURE" and "TAGS" sections below).

CARD READER:

00 ST LH LL <DATA: VAR>

EXAMPLE:

Let's say, for example, a transaction has been made and it has STAN (number of the transaction) = 326. Then we would like to get the following information for this specific transaction: transaction result, payment interface, contactless card scheme and the amount of the transaction. From the table below (TAGS), we can see, that we need to get tags 0xDF05, 0xDF25, 0xDF60 and 0x81. So in this case we need to send the following command:

(EXT DEVICE -> CARD READER) >>	3E 3D 00 00 0C 24 00 00 01 46 DF 05 DF 25 DF 60 81 72
(CARD READER -> EXT DEVICE) <<	3E 00 00 00 19 DF 05 04 00 00 00 00 DF 25 02 00 01 DF 60 04 00 00 00 03 81 04 00 00 00 64 59

CLARIFICATION:

(EXT DEVICE -> CARD READER):

'>':	0x3E - Start of the packet.
CMD:	0x3D - BORICA CMD command.
00:	0x00 - Fixed value '00'.
LH LL:	0x00 0x0C - The data length is 12 bytes.
SUBCMD:	0x24 - GET REPORT TAGS BY STAN subcommand.
<STAN: 4>:	'00 00 01 46' - HEX format (BIG ENDIAN), STAN = 326
<TAGS: VAR>:	'DF 05 DF 25 DF 60 81' - HEX format, here we need to list all tags, which we need to get for transaction with STAN = 326. In this case we want to get tags 0xDF05, 0xDF25, 0xDF60 and 0x81
CSUM:	0x72 - Xor of all bytes (3E 3D 00 ..... 60 81).

(CARD READER -> EXT DEVICE):

'>':	0x3E - Start of the packet.
00:	0x00 - Fixed value '00'.
ST:	0x00 - No error (ERROR CODES).
LH LL :	0x00 0x19 - The data length is 25 bytes.

<DATA: VAR>:	'DF 05 04 00 00 00 00 DF 25 02 00 01 DF 60 04 00 00 00 03 81 04 00 00 00 64' - HEX format, this is the information, which is returned for transaction with STAN = 326. It contains listed multiple elements, composed of <TAG, LEN, VALUE>. 'DF 05 (TAG) 04 (LEN) 00 00 00 00 (VALUE)' - Transaction result tag has value 0, which means the transaction was successful. 'DF 25 (TAG) 02 (LEN) 00 01 (VALUE)' - Payment interface has value 1. From the reference below, we can see, that this transaction was made with contactless interface. 'DF 60 (TAG) 04 (LEN) 00 00 00 03 (VALUE)' - Contactless card cheme has value 3. From the reference below, we can see, that this transaction was made with Visa Contactless card. '81 (TAG) 04 (LEN) 00 00 00 64 (VALUE) - The amount of this transaction is 100 (1.00).
CSUM:	0x59 - Xor of all bytes (3E 00 00 ..... 00 64).

Note: If some of the tags, listed in the command (EXT DEVICE -> CARD READER) is missing for this transaction, it will be missing in the response. If all of the tags are missing for this transaction, the response will contain ST field with value errNoData (8) error.

20. 0x25: SELECT CHIP APPLICATION

DESCRIPTION:

This command needs to be used only after receiving the event “0x3F: SELECT CHIP APPLICATION”. With this command the EXT DEVICE selects a chip application, where the index is selected by the user (indexes start from ‘00’ for the first application in the TLV format data from the event).

EXT DEVICE:

3D 00 LH LL 25 [CHIP APPLICATION INDEX: 1 BYTE]

CARD READER:

00 ST 00 00

EXAMPLE:

(EXT DEVICE -> CARD READER) >>	3E 3D 00 00 02 25 00 24
(CARD READER -> EXT DEVICE) <<	3E 00 00 00 00 3E

CLARIFICATION:

(EXT DEVICE -> CARD READER):

'>':	0x3E - Start of the packet.
CMD:	0x3D – Borica command.
00:	0x00 - Fixed value '00'.
LH LL:	0x00 0x02 - The data length is 2 bytes.
SUBCMD:	0x25 – SELECT CHIP APPLICATION subcommand.
[PASSWORD TYPE]	0x00 – Select chip app with index = 0 (the first one).
CSUM:	0x24 - Xor of all bytes (3E 3D 00 00 02 25 00).

(CARD READER -> EXT DEVICE)

'>':	0x3E - Start of the packet.
00:	0x00 - Fixed value '00'.
ST:	0x00 - errNoErr (ERROR CODES)
LH LL :	0x00 0x00 - The data length is 0 bytes.
CSUM:	0x3E - Xor of all bytes (3E 00 00 00 00).

21. 0x26: GET CARD READER STATE

EXT DEVICE:

3D 00 LH LL 26

CARD READER:

00 ST 00 01 [CARD READER STATE: 1 BYTE]

[CARD READER STATE: 1 BYTE]

0x01 – Idle mode
0x02 – Transaction is started
0x03 – Select application
0x04 – PIN entry
0x05 – Online authorization

EXAMPLE:

(EXT DEVICE -> CARD READER) >>	3E 3D 00 00 01 26 24
(CARD READER -> EXT DEVICE) <<	3E 00 00 00 01 01 3E

CLARIFICATION:

(EXT DEVICE -> CARD READER):

'>':	0x3E - Start of the packet.
CMD:	0x3D – Borica command.
00:	0x00 - Fixed value '00'.
LH LL:	0x00 0x01 - The data length is 1 byte.
SUBCMD:	0x26 – GET CARD READER STATE subcommand.
CSUM:	0x24 - Xor of all bytes (3E 3D 00 00 01 26).

(CARD READER -> EXT DEVICE)

'>':	0x3E - Start of the packet.
00:	0x00 - Fixed value '00'.
ST:	0x00 - errNoErr (ERROR CODES)
LH LL :	0x00 0x01 - The data length is 1 byte.
[CARD READER STATE]:	0x01 – Card reader is in IDLE mode.
CSUM:	0x3E - Xor of all bytes (3E 00 00 00 01 01).

22. 0x27: GET TERMINAL TAGS

DESCRIPTION:

Usually in order to get some information using the command “0x04: GET REPORT TAGS” the card reader needs to currently have Sale transactions in the log. If there are no transactions, no result will be returned. For this reason we introduced command “0x27: GET TERMINAL TAGS”, which will return few Terminal-related tags (Not Transaction-related) even if currently there are no Sale transactions in the log. This kind of important information are for example tags 0x9F1C (TERMINAL_ID), 0x9F1E (TERMINAL_SERIAL_NUMBER), 0xDF32 (BATCH_NUMBER) and many more.

EXT DEVICE:

3D 00 LH LL 27 <TAGS: VAR>

<TAGS: VAR>:

Tags to be returned. (For reference see "TLV STRUCTURE" and "TAGS" sections below).

CARD READER:

00 ST LH LL <DATA: VAR>

<DATA: VAR>:

Returned tags data

EXAMPLE:

(EXT DEVICE -> CARD READER) >>	3E 3D 00 00 05 27 9F 1E DF 32 4D
(CARD READER -> EXT DEVICE) <<	3E 00 00 00 12 9F 1E 0A 31 39 30 31 39 30 30 30 31 31 DF 32 02 00 64 2C

CLARIFICATION:

(EXT DEVICE -> CARD READER):

'>':	0x3E - Start of the packet.
CMD:	0x3D – Borica command.
00:	0x00 - Fixed value '00'.
LH LL:	0x00 0x05 - The data length is 5 bytes.
SUBCMD:	0x27 – GET TERMINAL TAGS subcommand.
<TAGS: VAR>	‘9F 1E DF 32’ - Here we need to list all tags, which we would like to get. In this case we need to get 2 tags – 0x9F1E and 0xDF32.
CSUM:	0x4D - Xor of all bytes (3E 3D 00 ..... DF 32).

(CARD READER -> EXT DEVICE)

'>':	0x3E - Start of the packet.
00:	0x00 - Fixed value '00'.
ST:	0x00 - No error (ERROR CODES).
LH LL :	0x00 0x12 - The data length is 18 bytes.

<DATA: VAR>:	'9F 1E 0A 31 39 30 31 39 30 30 30 31 31 DF 32 02 00 64' – Tag 0x9F1E (Serial number) has value “1901900011” and DF32 (Batch number) has value 100.
CSUM:	0x2C - Xor of all bytes (3E 00 00 ..... 00 64).

3. 0x40: External internet subcommands

External internet command contains multiple subcommands, which are associated mainly with handling the communication with the host. If the card reader does not use its own WIFI or GSM capability for communicating with the host, then the card reader will use the external device for this communication. The external device needs to implement mainly 4 different activities – opening a socket, closing a socket, sending data to the host and receiving response from the host and forwarding the response to the card reader. In order for this functionality to be working correctly, “External internet command” and “External internet event” need to be implemented and handled correctly.

1. 0x01: RECEIVE DATA

DESCRIPTION:

This command is being used when the external device receives some data from the host. Using this command, the EXT DEVICE sends the received data to the CARD READER.

EXT DEVICE:

40 00 LH LL [SUBCMD] <DATA: VAR>

<DATA>: The data, which is received from the host.

CARD READER:

00 ST 00 00

EXAMPLE:

(EXT DEVICE -> CARD READER) >>	3E 40 00 00 3C 01 14 03 01 00 01 01 16 03 01 00 30 17 D3 84 B3 F7 DF 99 20 A6 DF E2 59 6D 99 3B 5C 05 15 D7 D9 51 A6 69 79 D9 37 5C 40 92 86 67 3B 19 70 0D B3 69 24 CF B7 7F E6 E9 35 34 7A A1 E9 A0
(CARD READER -> EXT DEVICE) <<	3E 00 00 00 00 3E

CLARIFICATION:

(EXT DEVICE -> CARD READER):

'>':	0x3E - Start packet.
CMD:	0x40 - EXTERNAL INTERNET command.
00:	0x00 - Fixed value '00'.
LH LL:	0x00 0x3C - The data length is 60 bytes.
SUBCMD:	0x01 - RECEIVE DATA subcommand.

<DATA: VAR>: '14 03 01 00 ..... 34 7A A1 E9' - This is data, which was received from the host. The EXT DEVICE doesn't need to understand these bytes, it must just send them to the CARD READER.

CSUM: 0xA0 - Xor of all bytes (3E 40 00 ..... A1 E9).

(CARD READER -> EXT DEVICE):

'>': 0x3E - Start packet.
00: 0x00 - Fixed value '00'.
ST: 0x00 - No error (ERROR CODES).
LH LL : 0x00 0x00 - The data length is 0 bytes.
CSUM: 0x3E - Xor of all bytes (3E 00 00 00 00).

Note: When sending the response from the host to the CARD READER, the EXT DEVICE needs to analyze the MTU value, which is currently used. By default the MTU is 0x0400 (1024 bytes). This means that if the length of the response from the host is bigger than the MTU, then the EXT DEVICE needs to "cut" the response to a smaller packets and send each of them separately to the CARD READER. So if the host returns 3000 bytes, then the EXT DEVICE needs to call the "0x01: RECEIVE DATA" subcommand 3 times – the first 2 calls should have parameter "LH LL" with value = '04 01' (1 byte "SUBCMD" + 0x400 bytes "<DATA: VAR>") and the 3rd call should have parameter "LH LL" with value = '03 B9' (1 byte "SUBCMD" + 0x3B8 bytes "<DATA: VAR>").

Note 2: When sending the "0x01: RECEIVE DATA" subcommand, the EXT DEVICE needs to analyze the ST field from the response, which is returned from the CARD READER!!! If the CARD READER returns field ST with value 0x26 (38 = errBusy), then the EXT DEVICE needs to wait 100 milliseconds and send the same data again, 2nd time! And this loop needs to continue while the CARD READER returns field ST with value = 0x26. If the CARD READER returns field ST with value 0x00 (errNoErr), then the CARD READER successfully received the data and it is ready to receive the next data, if there is any.

2. 0x02: EVENT CONFIRM

DESCRIPTION:

This command confirms to the CARD READER whether the EXT DEVICE successfully performed an action, which before that was issued from the CARD READER with any of the corresponding events (for example open socket).

EXT DEVICE:

40 00 LH LL [SUBCMD] [TYPE] [RESULT] <MTU EXT DEVICE: 2>

[TYPE]:

[01]: SOCKET OPEN EVENT – Needs to be sent only if the CARD READER before that sent the “01: SOCKET OPEN” event.

[02]: SOCKET CLOSE EVENT – Needs to be sent only if the CARD READER before that sent the “02: SOCKET CLOSE” event.

[03]: SEND DATA EVENT – Needs to be sent only if the CARD READER before that sent the “03: SEND DATA” event.

[RESULT]:
[00]: CONFIRM OK
[01]: CONFIMR FAIL

<MTU EXT DEVICE>:
[MTU HIGH][MTU LOW]: Maximum transmit unit of the external device. By default it is 0x0400 (1024 bytes).

CARD READER:
00 ST LH LL

EXAMPLE:
(EXT DEVICE -> CARD READER) >> 3E 40 00 00 03 02 02 00 7D
(CARD READER -> EXT DEVICE) << 3E 00 00 00 00 3E

CLARIFICATION:
(EXT DEVICE -> CARD READER):
'>': 0x3E - Start packet.
CMD: 0x40 - EXTERNAL INTERNET command.
00: 0x00 - Fixed value '00'.
LH LL: 0x00 0x03 - The data length is 3 bytes.
SUBCMD: 0x02 - EVENT CONFIRM subcommand.
[TYPE]: 0x02 - SOCKET CLOSE EVENT
[RESULT]: 0x00 - CONFIRM OK
<MTU EXT DEVICE: 2>: Not present in this example.
CSUM: 0x7D - Xor of all bytes (3E 40 00 ..... 02 00).

(CARD READER -> EXT DEVICE):
'>': 0x3E - Start packet.
00: 0x00 - Fixed value '00'.
ST: 0x00 - No error (ERROR CODES).
LH LL : 0x00 0x00 - The data length is 0 bytes.
CSUM: 0x3E - Xor of all bytes (3E 00 00 00 00).

3. 0x03: GET MAX MTU

EXT DEVICE:
40 00 LH LL [SUBCMD]

CARD READER:
00 ST LH LL <MTU PINPAD: 2>

<MTU PINPAD>:
[MTU HIGH][MTU LOW]: Maximum transmit unit of the CARD READER. The card reader only reports its max MTU (how much bytes

maximum can be transferred at once between EXT DEVICE and the CARD READER using the subcommand “0x01: RECEIVE DATA”).

EXAMPLE:

(EXT DEVICE -> CARD READER) >> 3E 40 00 00 01 03 7C
(CARD READER -> EXT DEVICE) << 3E 00 00 00 02 04 00 38

CLARIFICATION:

(EXT DEVICE -> CARD READER):

'>':	0x3E - Start packet.
CMD:	0x40 - EXTERNAL INTERNET command.
00:	0x00 - Fixed value '00'.
LH LL:	0x00 0x01 - The data length is 1 byte.
SUBCMD:	0x03 - GET MAX MTU subcommand.
CSUM:	0x7C - Xor of all bytes (3E 40 00 ..... 01 03).

(CARD READER -> EXT DEVICE):

'>':	0x3E - Start packet.
00:	0x00 - Fixed value '00'.
ST:	0x00 - No error (ERROR CODES).
LH LL :	0x00 0x02 - The data length is 2 bytes.
<MTU PINPAD: 2>:	0x04 0x00 - MAX MTU is 0x400 bytes.
CSUM:	0x38 - Xor of all bytes (3E 00 00 ..... 04 00).

4. 0x0E: Borica subevents

At some point different events can be received from the CARD READER. The Borica event has value 0x0E and the different Borica subevents are mostly connected with information for the transaction, which is currently performed, or a transaction, which was already performed in the past.

1. 0x01: TRANSACTION COMPLETE (PAYMENT RESULT) - NO CONFIRMATION NEEDED

CARD READER:

0E 00 LH LL 01 <DATA: VAR>

<DATA>:

Tags, encoded in TLV format:

BOR_TAG_PAY_RES	0xDF05 - [HEX BIG ENDIAN]
BOR_TAG_PAY_ERR	0xDF06 - [HEX BIG ENDIAN]
BOR_TAG_HOST_RRN	0xDF07 - [STRING]
BOR_TAG_HOST_AUTH_ID	0xDF08 - [STRING]
TAG_81_AMOUNT_AUTHORISED_BINARY	0x0081 - [HEX BIG ENDIAN]

BOR_TAG_EMV_STAN	0x9F41 - [BCD]
BOR_TAG_MAX_CASHBACK_AMOUNT	0xDF8004 - [HEX BIG ENDIAN]
BOR_TAG_CASHBACK_CURRENCY	0xDF8005 - [STRING]
BOR_TAG_BCARD_SCA_DECLINED	0xDF8006 - [HEX BIG ENDIAN]

EXAMPLE:

(CARD READER -> EXT DEVICE) >> 3E 0E 00 00 1C 01 DF 05 04 00 00 00 00 DF 06 04 00 00 00 00 81 04 00 00 00 00 9F 41 04 00 00 03 67 15

CLARIFICATION:

(CARD READER -> EXT DEVICE):

'>':	0x3E - Start of the packet.
EVENT:	0x0E - Borica event.
00:	0x00 - Fixed value '00'.
LH LL:	0x00 0x1C - The data length is 28 bytes.
SUBEVENT:	0x01 - TRANSACTION COMPLETE subevent.
<DATA: VAR>:	'DF 05 04 00 00 00 00 DF 06 04 00 00 00 00 81 04 00 00 00 00 9F 41 04 00 00 03 67' - The Result (Tag 0xDF05) is 0 (Success), the Error (Tag 0xDF06) is 0 (No Error), The Amount (Tag 0x81) is 0 and the transaction number (Tag 0x9F41) is 367.
CSUM:	0x15 - Xor of all bytes (3E 0F 00 ..... 4E EE).

2. 0x02: INTERMEDIATE TRANSACTION COMPLETE (PAYMENT RESULT) - NO CONFIRMATION NEEDED

Sent if a hung transaction passed just before the transaction, which is currently being executed.

CARD READER:

0E 00 LH LL 02 <DATA: VAR>

<DATA>:

Tags, encoded in TLV format:

BOR_TAG_PAY_RES	0xDF05 - [HEX BIG ENDIAN]
BOR_TAG_PAY_ERR	0xDF06 - [HEX BIG ENDIAN]
BOR_TAG_HOST_RRN	0xDF07 - [STRING]
BOR_TAG_HOST_AUTH_ID	0xDF08 - [STRING]
TAG_81_AMOUNT_AUTHORISED_BINARY	0x0081 - [HEX BIG ENDIAN]
BOR_TAG_EMV_STAN	0x9F41 – [BCD]

3. 0x03: PRINT HANG TRANSACTION RECEIPT

DESCRIPTION:

If this Event is received, a receipt for a Hang transaction should be force printed, containing all required information, which is included in the Event.

CARD READER:

0E 00 LH LL 03 <DATA: VAR>

<DATA>:

Needed tags for constructing a receipt (See 7. TAGS).

EXAMPLE:

(CARD READER -> EXT DEVICE) >> 3E 0E 00 01 EC 03 9F 1C 08 39 33 38 30 30 34 37 35 9F 16 0F 39 39 39 33 39 30 30 30 30 31 30 30 30 30 30 5F 2A 02 09 75 DF 26 02 0A 02 DF 27 03 42 47 4E DF 28 07 39 37 30 32 36 31 30

..... (too long data) .....

81 04 00 00 00 05 9F 41 04 00 00 17 79 DF 07 0C 32 31 39 39 30 36 37 35 37 33 37 35 DF 08 06 30 35 31 32 39 36 3F

CLARIFICATION:

(CARD READER -> EXT DEVICE):

'>':	0x3E - Start of the packet.
EVENT:	0x0E - Borica event.
00:	0x00 - Fixed value '00'.
LH LL:	0x01 0xEC - The data length is 492 bytes.
SUBEVENT:	0x03 - PRINT HANG TRANSACTION RECEIPT subevent.
<DATA: VAR>:	' 9F 1C 08 39 33 38 30 30 34 37 ... 31 32 39 36 35' – Data in TLV format (See 7. TAGS). This information should be used for constructing and printing a receipt.
CSUM:	0x3F - Xor of all bytes (3E 0E 00 ..... 39 36).

4. 0x10: SEND LOG DATA

Sent as a result of subcommand "0x04: GET REPORT TAGS" with parameter "00 10 - GET ALL RECORDS AS EVENTS". Every record is sent as a single event.

CARD READER:

0E 00 LH LL 10 <DATA: VAR>

5. 0x3F: SELECT CHIP APPLICATION

DESCRIPTION:

This event will be sent only from BlueCash50 device! This event will be sent only if transaction is performed with chip card, which supports more than one chip applications. This event contains the names of the applications, which are supported from the card, they need to be displayed in a menu and after the selection, the EXT DEVICE needs to send command "0x25: SELECT CHIP APPLICATION" in order to confirm which is the selected application by the user.

CARD READER:

0E 00 LH LL 3F <DATA: VAR>

<DATA>:

The names of the applications, encoded in TLV structure

BOR_TAG_CHIP_APPLICATION_NAME 0xDFC101 - [STRING]

EXAMPLE:

(CARD READER -> EXT DEVICE) >> 3E 0E 00 00 21 3F DF C1 01 0C 43 6F 6D 62 6F 30 31 20 76 31 20 31 DF C1 01 0C 43 6F 6D 62 6F 30 31 20 76 31 20 31 2E

CLARIFICATION:

(CARD READER -> EXT DEVICE):

'>':	0x3E - Start of the packet.
EVENT:	0x0E - Borica event.
00:	0x00 - Fixed value '00'.
LH LL:	0x00 0x21 - The data length is 33 bytes.
SUBEVENT:	0x3F - SELECT CHIP APPLICATION subevent.
<DATA: VAR>:	'DF C1 01 0C 43 6F 6D 62 6F 30 31 20 76 31 20 31 DF C1 01 0C 43 6F 6D 62 6F 30 31 20 76 31 20 31' – TLV format – Each tag 0xDFC101 contains the name of a specific application. They need to be displayed in the same order and they have indexes starting from ‘00’ for the first application.

5. 0x0F: External internet subevents

At some point different events can be received from the CARD READER. The External internet event has value 0x0F and the different External internet subevents are mostly connected with the communication with the host – opening/closing a socket or sending data to the host.

1. 0x01: SOCKET OPEN

DESCRIPTION:

When the EXT DEVICE receives this subevent, it must open socket to <ADDRESS>:<PORT> with <TIMEOUT> and if it is using APN SIM card, it needs to use the APN, Username and Password (From the tags below) for the SIM card. Then the EXT DEVICE needs to send the subcommand "2. 0x02: EVENT CONFIRM".

CARD READER:

0F 00 LH LL [SUBEVENT] [ID] [TYPE] <ADDRESS: 4> <PORT: 2> <TIMEOUT: 2> <DATA: VAR>

[ID]:
- [0..255]: SOCKET ID

[TYPE]:
- [01]: TCP
- [02]: UDP
- [03]: TCP TLS
- [04]: UDP TLS

<ADDRESS>:
- [0..255][0..255][0..255][0..255] : SOCKET ADDRESS

<PORT>:
- [PORT HIGH][PORT LOW] : SOCKET PORT

<TIMEOUT>:
- [TIMEOUT HIGH][TIMEOUT LOW] : TIMEOUT IN SECONDS

<DATA>:
- TAGS ENCODED IN TLV FORMAT (For reference see "TLV STRUCTURE" section below):

BOR_TAG_COMM_APN	0xDF50 - [STRING]
BOR_TAG_COMM_GPRS_UN	0xDF53 - [STRING]
BOR_TAG_COMM_GPRS_PWD	0xDF54 - [STRING]
BOR_TAG_HOST_LAN_IP	0xDF6C - [STRING]
BOR_TAG_HOST_LAN_PORT	0xDF6D - [HEX]

EXAMPLE:

(CARD READER -> EXT DEVICE) >> 3E 0F 00 00 26 01 01 01 55 FF 80 C4 4E ED 00 1E DF 50 06 62 6F 72 69 63 61 DF 53 00 DF 54 00 DF 6C 04 58 CB EF 2E DF 6D 02 4E EE 2A

CLARIFICATION:

(CARD READER -> EXT DEVICE):

'>':	0x3E - Start packet.
EVENT:	0x0F - EXTERNAL INTERNET event.
00:	0x00 - Fixed value '00'.
LH LL:	0x00 0x26 - The data length is 38 bytes.
SUBEVENT:	0x01 - SOCKET OPEN subevent.
[ID]:	0x01 - Socket ID
[TYPE]:	0x01 - TCP
<ADDRESS: 4>:	0x55 0xFF 0x80 0xC4 - 85.255.128.196

<PORT: 2>: 0x4E 0xED - 20205
<TIMEOUT: 2>: 0x00 0x1E – 30

<DATA: VAR>: 'DF 50 06 62 6F 72 69 63 61 DF 53 00 DF 54 00 DF 6C 04 58 CB EF 2E DF 6D 02 4E EE' - The APN (Tag 0xDF50) is "borica", the Username (Tag 0xDF53) is "" (none) and the Password (Tag 0xDF54) is also "". The other tags don't need to be used.

CSUM: 0x2A - Xor of all bytes (3E 0F 00 ..... 4E EE).

2. 0x02: SOCKET CLOSE

DESCRIPTION:

When the EXT DEVICE receives this subevent, it must close the socket with [ID], which was opened before. Then the EXT DEVICE needs to send the subcommand "2. 0x02: EVENT CONFIRM".

CARD READER:

0F 00 LH LL [SUBEVENT] [ID]

[ID]:

[0..255]: SOCKET ID

EXAMPLE:

(CARD READER -> EXT DEVICE) >> 3E 0F 00 00 02 02 01 30

CLARIFICATION:

(CARD READER -> EXT DEVICE):

'>': 0x3E - Start packet.
EVENT: 0x0F - EXTERNAL INTERNET event.
00: 0x00 - Fixed value '00'.
LH LL: 0x00 0x02 - The data length is 2 bytes.
SUBEVENT: 0x02 - SOCKET CLOSE subevent.
[ID]: 0x01 - ID of the socket = 1.
CSUM: 0x30 - Xor of all bytes (3E 0F 00 ..... 02 01).

3. 0x03: SEND DATA

DESCRIPTION:

When the EXT DEVICE receives this subevent, it needs to transmit the <DATA> to the host. Then the EXT DEVICE needs to send the subcommand "2. 0x02: EVENT CONFIRM".

CARD READER:

0F 00 LH LL [SUBEVENT] [ID] <DATA: VAR>
[ID]:

[0..255]: SOCKET ID

<DATA>:

DATA TO SEND TO THE SOCKET

EXAMPLE:

(CARD READER -> EXT DEVICE) >> 3E 0F 00 00 37 03 01 16 03 01 00 30 A9 1C C0 83
3E 74 95 D5 84 3B BE 6E D2 06 DE B2 B4 65 1E
75 CE 43 60 DC 6E 26 45 EE C7 B1 7C 2D 1E A1
E3 ED 9B A1 7F 18 BD 30 24 30 9F C1 75 72 68

CLARIFICATION:

(CARD READER -> EXT DEVICE):

'>':	0x3E - Start packet.
EVENT:	0x0F - EXTERNAL INTERNET event.
00:	0x00 - Fixed value '00'.
LH LL:	0x00 0x37 - The data length is 55 bytes.
SUBEVENT:	0x03 - SEND DATA subevent.
[ID]:	0x01 - ID of the socket is 1.
<DATA: VAR>:	'16 03 01 00 ..... C1 75 72' - The EXT DEVICE doesn't need to understand these bytes, it must just send them to the host.
CSUM:	0x68 - Xor of all bytes (3E 0F 00 ..... 75 72).

6. ERROR CODES

errNoErr	0	no error
errGeneral	1	some error
errInvCmd	2	no valid command or subcommand code
errInvPar	3	invalid paremeter
errInvAdr	4	address is outside limits
errInvVal	5	value is outside limits
errInvLen	6	length is outside limiuts
errNoPermit	7	the action is not permit in current state
errNoData	8	there is no data to be returned
errTimeOut	9	timeout is occur
errKeyNum	10	invalid key number
errKeyAttr	11	invalid key attributes(ussage)
errKeyUsage	11	invalid key attributes(ussage)
errInvDevice	12	calling of non-existing device
errNoSupport	13	(no used in this FW version)
errPinLimit	14	Pin entering limit exceed
errFlash	15	some error in flash commands
errHard	16	some hardware error
errInvCRC	17	(no used in this FW version)
errCancel	18	the button "CANCEL" is pressed
errInvSign	19	invalid signature
errInvHead	20	invalid data in header
errInvPass	21	incorrent password
errKeyFormat	22	invalid key format

errSCR	23	some error in smart card reader
errHAL	24	error code is returned from HAL functions
errInvKey	25	invalid key (may be no pressent)
errNoPinData	26	the PIN length is <4 or >12
errInvRemainder	27	Issuer or ICC key invalid remainder length
errNoInit	28	(no used in this FW version)
errLimit	29	(no used in this FW version)
errInvSeq	30	(no used in this FW version)
errNoPerm	31	The action is not permited
errNoTMK	32	TMK is not loaded. The action cannot be executed
errInvKek	33	Wrong kek format
errDubKey	34	Dublicated key
errKBD	35	general keyboard error
errKBDNoCal	36	the keyboard is no calibrated
errKBDBug	37	keyboard bug detected
errBusy	38	the device is busy, tray again
errTampered	39	the devicete in security tampered
errEmsr	40	error in encrypted head
errAccept	41	the button "OK" is pressed
errInvPAN	42	wrong PAN
errOutOfMemory	43	no enought memory
errEMV	44	some EMV error
errCrypt	45	cryptographic error
errComRcv	46	something is received
errWrongVer	47	wrong version. update to old firmware for example
errNoPaper	48	printer is out of paper
errTooHot	49	printer is overheating
errNoConnected	50	the device is not connected
errUseChip	51	use the chip reader
errEndDay	52	end the day first

7. TAGS

[HEX]	TAG_DF05_TRANSACTION_RESULT	0xDF05	REF. 1)
[HEX]	TAG_DF06_TRANSACTION_ERROR_CODE	0xDF06	REF. "ERROR CODES"
[STRING]	TAG_DF31_TERMINAL_MERCHANT_NAME_BG	0xDF31	
[STRING]	TAG_DF30_TERMINAL_MERCHANT_CITY_BG	0xDF30	
[STRING]	TAG_DF2F_TERMINAL_MERCHANT_ADDRESS_BG	0xDF2F	
[STRING]	TAG_DF29_TERMINAL_MERCHANT_POST_CODE	0xDF29	
[STRING]	TAG_DF2E_TERMINAL_MERCHANT_TITLE_BG	0xDF2E	
[STRING]	TAG_DF28_TERMINAL_MERCHANT_PHONE	0xDF28	
[STRING]	TAG_9F1C_TERMINAL_ID	0x9F1C	
[STRING]	TAG_DF00_CARD_SCHEME	0xDF00	
[STRING]	TAG_DF0A_MASKED_PAN	0xDF0A	
[STRING]	TAG_5F20_CARDHOLDER_NAME	0x5F20	
[HEX]	BOR_TAG_PAY_INTERFACE	0xDF25	REF. 2)
[HEX]	BOR_TAG_CARD_SCHEME_CL	0xDF60	REF. 3)
[HEX]	TAG_DF10_TRANS_TYPE	0xDF10	REF. 4)
[HEX]	TAG_DF0B_LOYALTY_PTS	0xDF0B	
[HEX]	TAG_81_AMOUNT_AUTHORISED	0x81	
[HEX]	TAG_DF04_AMOUNT_EUR	0xDF04	

[HEX] TAG_DF23_CVM_SIGNATURE	0xDF23	REF. 5)
[BCD] TAG_9F41_TRANSACTION_SEQUENCE_COUNTER	0x9F41	
[STRING] TAG_DF07_HOST_RRN	0xDF07	
[STRING] BOR_TAG_HOST_AUTH_ID	0xDF08	
[HEX] BOR_TAG_HOST_CODE	0xDF09	
[HEX] BOR_TAG_TRANS_BATCH_NUM	0xDF61	
[STRING] TAG_9F16_TERMINAL_MERCHANT_ID	0x9F16	
[STRING] BOR_TAG_INTERFACE_ID	0xDF62	
[BCD] TAG_9F06_TERM_AID	0x9F06	
[HEX] TAG_9F26_APP_CRYPTOGRAM	0x9F26	
[BCD] TAG_9A_TRANSACTION_DATE	0x9A	[Year:Month:Day]
[BCD] TAG_9F21_TRANSACTION_TIME	0x9F21	[Hour:Minute:Seconds]
[HEX] BOR_TAG_ISSUER_ID	0xDF79	REF. 6)
[HEX] BOR_TAG_TO_TYPE	0xDF12	REF. 7)
[STRING] BOR_TAG_TERM_CURR_NAME	0xDF27	
[HEX] BOR_CMD_TAG_CASHBACK	0x9F04	
[HEX] BOR_TAG_BCARD_CID_1	0xDF7D	
[HEX] BOR_TAG_BCARD_CID_2	0xDF7E	
[HEX] BOR_TAG_IS_PIN_ENTERED	0xDF7F	
[STRING] BOR_TAG_CHIP_APPLICATION_NAME	0xDFC101	
[HEX] BOR_TAG_MAX_CASHBACK_AMOUNT	0xDF8004	
[STRING] BOR_TAG_CASHBACK_CURRENCY	0xDF8005	
[HEX] BOR_TAG_BCARD_SCA_DECLINED	0xDF8006	
[STRING] BOR_TAG_TERMINAL_SERIAL_NUMBER	0x9F1E	
[HEX] BOR_TAG_BATCH_NUMBER	0xDF32	

8. REFERENCES

1. TRANSACTION_RESULTS

```
/** @brief Transaction is approved.*/
#define PAY_TRANSACTION_APPROVED                0
/** @brief Transaction is declined.*/
#define PAY_TRANSACTION_DECLINED                1
/** @brief Transaction error.*/
#define PAY_TRANSACTION_ERROR                   2
/** @brief Try other interface. Contactless only.*/
#define PAY_TRANSACTION_TRY_OTHER_IF            3
/** @brief Try again. Contactless only.*/
#define PAY_TRANSACTION_TRY_AGAIN               4
```

2. INTERFACES

```
/** @brief Chip interface is used.*/
#define PAY_INTERFACE_CHIP                      0
/** @brief Contactless interface is used.*/
#define PAY_INTERFACE_CONTACTLESS               1
/** @brief Magnetic stripe interface is used.*/
#define PAY_INTERFACE_MAG_STRIPE                2
/** @brief Manual card entry is used.*/
#define PAY_INTERFACE_MANUAL_ENTRY              3
```

3. CL CARD SCHEMES

```
/** @brief Contactless Card Scheme IDs. */
#define PAY_CL_CARD_SCHEME_VISA_AP          0x01
#define PAY_CL_CARD_SCHEME_PAYPASS          0x02
#define PAY_CL_CARD_SCHEME_VISA             0x03
#define PAY_CL_CARD_SCHEME_AMEX             0x04
#define PAY_CL_CARD_SCHEME_JCB              0x05
#define PAY_CL_CARD_SCHEME_DISCOVER         0x06
#define PAY_CL_CARD_SCHEME_BCARD            0x0A
```

4. TRANSACTION TYPES

```
#define BOR_CMD_TRANS_TYPE_PURCH            1
#define BOR_CMD_TRANS_TYPE_PURCH_CASH       2
#define BOR_CMD_TRANS_TYPE_PURCH_REF        3
#define BOR_CMD_TRANS_TYPE_CASH             4
#define BOR_CMD_TRANS_TYPE_AUTH             5
#define BOR_CMD_TRANS_TYPE_PURCH_CODE       6
#define BOR_CMD_TRANS_TYPE_VOID_PURCH       7
#define BOR_CMD_TRANS_TYPE_VOID_CASH        8
#define BOR_CMD_TRANS_TYPE_VOID_AUTH        9
#define BOR_CMD_TRANS_TYPE_END_OF_DAY       10
#define BOR_CMD_TRANS_TYPE_LOYALTY_BALANCE  11
#define BOR_CMD_TRANS_TYPE_LOYALTY_SPEND    12
#define BOR_CMD_TRANS_TYPE_VOID_LOYALTY_SPEND 13
#define BOR_CMD_TRANS_TYPE_TEST_CONN        14
#define BOR_CMD_TRANS_TYPE_TMS_UPDATE       15
```

5. SIGNATURE TYPES

```
#define BOR_CVM_SIGN_CARDHOLDER             1
#define BOR_CVM_SIGN_MERCHANT               2
```

6. CARD ISSUERS

```
#define BOR_ISS_ID_BORICA                   1
#define BOR_ISS_ID_VISA                     2
#define BOR_ISS_ID_MASTERCARD               3
#define BOR_ISS_ID_MAESTRO                  4
#define BOR_ISS_ID_AMEX                     5
#define BOR_ISS_ID_DINERS_CLUB              6
#define BOR_ISS_ID_JCB                      7
#define BOR_ISS_ID_AMEX_PIN                 8
#define BOR_ISS_ID_BCARD                    9
```

7. TO TYPES

```
#define BOR_TO_TYPE_NONE        0
#define BOR_TO_TYPE_REVERSAL    1
#define BOR_TO_TYPE_HANG        2
```

9. TLV STRUCTURE

When a transaction is performed, some information about it is being saved in the memory of the CARD READER. This information is array, composed of multiple elements - tags in TLV format. TLV (Tag - Len - Value) contains the name of the tag, length of the data and the data. For example let's say Purchase was performed. Its amount is 1.00 BGN. So when this information is saved in TLV format it will look like this:

81 04 00 00 00 64

- 81 - This is the Tag (TAG_81_AMOUNT_AUTHORISED - 0x81)
- 04 - This is the Length of the data. It means, that the next 4 bytes will be the amount of this transaction.
- 00 00 00 64 - This is the Data for tag 0x81. It is in HEX format and it means that the amount = 100 (1.00)

Now let's say the Purchase, which was performed, has more information, which is saved in the memory. It can look like this:

DF 10 02 00 01 81 04 00 00 00 64 DF 04 04 00 00 00 33 DF 23 02 00 01 DF 25 02 00 00 DF 61 02 00 73 DF 00 0B 56 69 73 61 20 43 72 65 64 69 74 5F 20 1A 56 49 53 41 20 41 43 51 55 49 52 45 52 20 54 45 53 54 2F 43 41 52 44 20 31 31 DF 62 02 43 31 9F 41 04 00 00 16 39 9F 21 03 09 34 22 9A 03 18 10 19 DF 09 04 00 00 00 00 DF 07 0C 38 32 39 32 30 36 36 38 33 30 33 33 DF 08 06 30 33 30 31 30 33 DF 05 04 00 00 00 00 DF 06 04 00 00 00 00 9F 06 08 A0 00 00 00 03 10 10 02

Online TLV parser (For example http://paymentcardtools.com/emv-tlv-parser) can be used, so that the array can easily be understood. When we parse the data, it has the following meanings:

DF 10 02 00 01

- DF 10 - Tag 0xDF10 (TAG_DF10_TRANS_TYPE)
- 02 - Lenght of the data = 2 bytes
- 00 01 - The value of transaction type. From "Reference 4" from above we can see, that this means, that the transaction is "Purchase".

81 04 00 00 00 64

- 81 - Tag 0x81 (TAG_81_AMOUNT_AUTHORISED)
- 04 - Length of the data = 4 bytes
- 00 00 00 64 - The value of "Amount" is 100 (1.00)

DF 04 04 00 00 00 33

- DF 04 - Tag 0xDF04 (TAG_DF04_AMOUNT_EUR)
- 04 - Length of the data = 4 bytes
- 00 00 00 33 - The value of "Amount EUR" is 51 (0.51)

DF 23 02 00 01

DF 23 - Tag 0xDF23 (TAG_DF23_CVM_SIGNATURE)

02 - Length of the data = 2 bytes

00 01 - The value of "Signature" is 1. From "Reference 5" from above we can see, that this means, that for this transaction Cardholder signature is needed.

DF 25 02 00 00

DF 25 - Tag 0xDF25 (BOR_TAG_PAY_INTERFACE)

02 - Length of the data = 2 bytes

00 00 - The value of "Payment Interface" is 0. From "Reference 2" from above we can see, that this transaction was performed with Chip interface.

DF 61 02 00 73

DF 61 - Tag 0xDF61 (BOR_TAG_TRANS_BATCH_NUM)

02 - Length of the data = 2 bytes

00 73 - This transaction was performed in batch with number = 115

DF 00 0B 56 69 73 61 20 43 72 65 64 69 74

DF 00 - Tag 0xDF00 (TAG_DF00_CARD_SCHEME)

0B - Lenght of the data = 11 bytes

56 69 73 61 20 43 72 65 64 69 74 - The card, which was used in this transaction, has card scheme = "Visa Credit".

5F 20 1A 56 49 53 41 20 41 43 51 55 49 52 45 52 20 54 45 53 54 2F 43 41 52 44 20 31 31

5F 20 - Tag 0x5F20 (TAG_5F20_CARDHOLDER_NAME)

1A - Length of the data = 26 bytes

56 49 53 41 20 41 43 51 55 49 52 45 52 20 54 45 53 54 2F 43 41 52 44 20 31 31 - The cardholder name is "VISA ACQUIRER TEST/CARD 11".

DF 62 02 43 31

DF 62 - Tag 0xDF62 (BOR_TAG_INTERFACE_ID)

02 - Length of the data = 2 bytes

43 31 - "C1"

9F 41 04 00 00 16 39

9F 41 - Tag 0x9F41 (TAG_9F41_TRANSACTION_SEQUENCE_COUNTER)

04 - Length of the data = 4 bytes

00 00 16 39 - This transaction is has number 1639.

9F 21 03 09 34 22

9F 21 - Tag 0x9F21 (TAG_9F21_TRANSACTION_TIME)

03 - Length of the data = 3 bytes

09 34 22 - [Hour:Minute:Seconds] The time of the transaction is 09:34:22 .

9A 03 18 10 19

9A - Tag 0x9A (TAG_9A_TRANSACTION_DATE)

03 - Length of the data = 3 bytes

18 10 19 - [Year:Month:Day] The date of the transaction is 19.10.2018.

DF 09 04 00 00 00 00

DF 09 -	Tag 0xDF09 (BOR_TAG_HOST_CODE)
04 -	Length of the data = 4 bytes
00 00 00 00 -	The host error code is 0, which means it is successful transaction (no error).

DF 07 0C 38 32 39 32 30 36 36 38 33 30 33 33

DF 07 -	Tag 0xDF07 (TAG_DF07_HOST_RRN)
0C -	The length of the data = 12 bytes
38 32 39 32 30 36 36 38 33 30 33 33 -	The unique RRN number for this transaction is "829206683033".

DF 08 06 30 33 30 31 30 33

DF 08 -	Tag 0xDF08 (BOR_TAG_HOST_AUTH_ID)
06 -	Length of the data = 6 bytes
30 33 30 31 30 33 -	The unique Authorization ID number for this transaction is "030103".

DF 05 04 00 00 00 00

DF 05 -	Tag 0xDF05 (TAG_DF05_TRANSACTION_RESULT)
04 -	Length of the data = 4 bytes
00 00 00 00 -	The transaction result has value 0. From "Reference 1" from above we can see, that this transaction is approved (PAY_TRANSACTION_APPROVED).

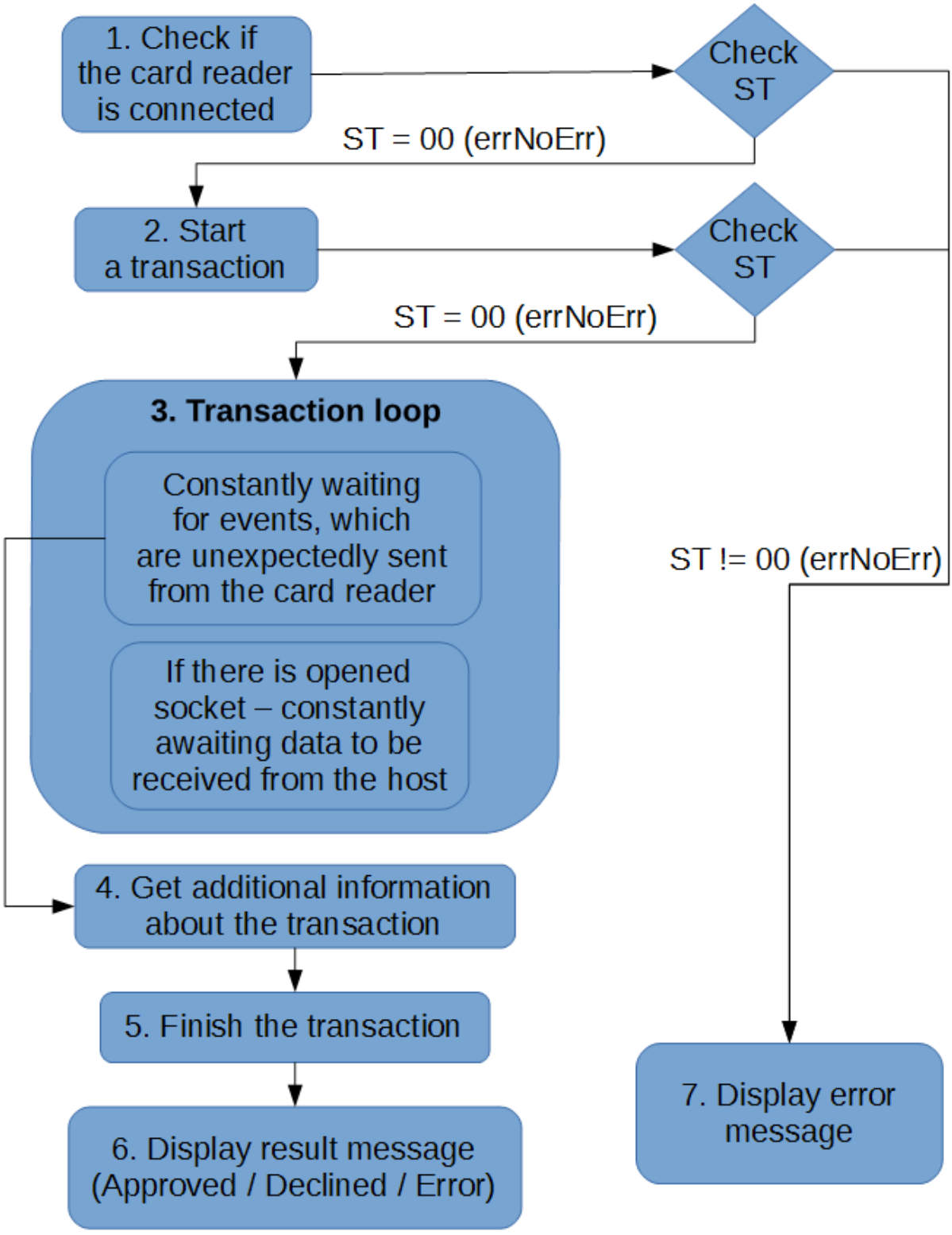
DF 06 04 00 00 00 00

DF 06 -	Tag 0xDF06 (TAG_DF06_TRANSACTION_ERROR_CODE)
04 -	Length of the data = 4 bytes
00 00 00 00 -	Transaction error code is 0 (there is no error).

9F 06 08 A0 00 00 00 03 10 10 02

9F 06 -	Tag 0x9F06 (TAG_9F06_TERM_AID)
08 -	Length of the data = 8 bytes
A0 00 00 00 03 10 10 02 -	The Application ID of the card, which was used in this transaction is A000000003101002, which means it is VISA card.

10. TRANSACTION FLOW



- **CLARIFICATIONS**

1. Check if the reader is connected

Send “PING” command and see what is the ST value in the response from the CARD READER.

2. Start a transaction

Send “TRANSACTION START” command and see what is the ST value in the response from the CARD READER. Different subcommands can be used for starting different types of transaction.

3. Transaction loop

Once the EXT DEVICE successfully starts a transaction (the CARD READER returns a response, containing ST field with value ‘00’), then the EXT DEVICE needs to “enter” in a transaction loop, where few things need to be constantly executed:

- At any moment the CARD READER can unexpectedly send an event, which is indicating that something needs to be performed. The EXT DEVICE needs to be capable at any moment to receive such an event and execute it.

  If the CARD READER has its own WIFI or GSM module, it will communicate with the host by itself and basically the only event, which will be sent, is “0x01: TRANSACTION COMPLETE” when the transaction is completed.

  If the CARD READER does NOT have its own capability for communicating with the host, then in the Transaction loop the EXT DEVICE needs to be able to also receive all events, which are described in the section “0x0F: External internet subevents”.

  Once the EXT DEVICE receives the event “0x01: TRANSACTION COMPLETE”, then it can “exit” from the Transaction loop and go to step 4.

  Note: If chip card with multiple chip applications is used, **only the BlueCash-50 device** at some point will also send event “0x3F: SELECT CHIP APPLICATION” and the EXT DEVICE should be able to execute it.

- If there is socket, which was previously opened from the EXT DEVICE (this happens only after the CARD READER sends the event “0x01: SOCKET OPEN” and before the CARD READER sends the event “0x02: SOCKET CLOSE”, when the socket is being closed), then the EXT DEVICE needs to constantly wait for data to be returned from the host at any time. Once any kind of data is returned from the host, it needs to be sent to the CARD READER using the subcommand “0x01: RECEIVE DATA”.

4. Get additional information

Once the transaction is completed (The event “0x01: TRANSACTION COMPLETE” was received from the CARD READER), then the EXT DEVICE can get additional information for the currently performed transaction by calling the subcommand “0x02: GET RECEIPT TAGS”.

5. Finish the transaction

At the end of the transaction, the EXT DEVICE needs to send to the CARD READER the subcommand “0x03: TRANSACTION END”.

6. Display result message

When the transaction is completed, the CARD READER will send to the EXT DEVICE the subevent “0x01: TRANSACTION COMPLETE”. This subevent will contain the following tags:

- 0xDF05 (Transaction result)
- 0xDF06 (Error code)
- 0xDF8004 (Maximum cashback amount) – In most cases this tag will be missing (optional).
- 0xDF8005 (Cashback currency) – In most cases this tag will be missing (optional).
- 0xDF8006 (Bcard SCA is declined) – In most cases this tag will be missing (optional).

When displaying the final result message (Approved / Declined / Error), the EXT DEVICE has to implement the following logic:

- If tag 0xDF05 has value ‘00’ (Approved), then the message should be “Transaction Approved” (or similar).
- Else if both tags 0xDF8004 and 0xDF8005 are present (With example value ‘DF 80 04 04 00 00 00 64 DF 80 05 03 42 47 4E’), then the following message should be displayed “Transaction declined! Maximum amount in cash is 100 BGN!”, where “100” is the example value of tag 0xDF8004 and “BGN” is the example value of tag 0xDF8005. These 2 tags can be sent only from the device BlueCash-50!
- Else if tag 0xDF8006 is present (With example value ‘DF 80 06 01 01’, the value will always be hardcoded to 01 here), the following message should be displayed “Transaction declined, PIN required! Please try again!”. This tag can be sent only from the device BlueCash-50!
- Else if tag 0xDF05 has value ‘01’ (Declined), then the message should be “Transaction Declined (<error_code>)”, or similar, where the <error code> is the value of tag 0xDF09 (Host status code), which can additionally be taken using the subcommand “0x02: GET RECEIPT TAGS”. Please check “13. HOST ERROR STRINGS”, if <error_code> value is defined there, an additional error string can be added to the message “Transaction Declined (<error_code>). <Error string>”, so for example if tag 0xDF09 has value ‘DF 09 04 00 00 00 37’, then the message would be “Transaction Declined (55). Incorrect PIN”.
- Else if tag 0xDF05 has value ‘02’ (Error) or in all other cases (by default), the message should be “Transaction error (<error_code>)”, or similar, where the <error_code> is the value of tag

0xDF06. Please check “12. CARD READER ERROR STRINGS”, if <error_code> value is defined there, an additional error string can be added to the message “Transaction error (<error_code>). <Error string>”, so for example if tag 0xDF06 has value ‘DF 06 04 00 00 00 12’, then the message would be “Transaction error (18). Operation canceled”.

7. Display error message

If there is any error before even starting the transaction, the EXT DEVICE should display correct error message.

11. EXAMPLE TRANSACTION

This is an example of the entire process that is needed to successfully execute a transaction.

1. The EXT DEVICE sends "0x00: PING" subcommand.

>> (EXT DEVICE): 3E 3D 00 00 01 00 02

<< (CARD READER): 3E 00 00 00 00 3E

2. The EXT DEVICE sends "0x01: TRANSACTION START" subcommand. This is an example for "Test Connection", but if you want to start a different transaction, you can see the examples, given in the subcommand.

>> (EXT DEVICE): 3E 3D 00 00 0D 01 0E 9A 03 19 03 26 9F 21 03 15 11 02 1F

<< (CARD READER): 3E 00 00 00 00 3E

3. At some point the CARD READER will need to open a socket. Then it will send the subevent "0x01: SOCKET OPEN".

<< (CARD READER): 3E 0F 00 00 26 01 01 01 55 FF 80 C4 4E ED 00 1E DF 50 06 62 6F 72 69 63 61 DF 53 00 DF 54 00 DF 6C 04 58 CB EF 2E DF 6D 02 4E EE 2A

4. The EXT DEVICE must confirm if it opened the socket successfully. It must send the subcommand "0x02: EVENT CONFIRM".

>> (EXT DEVICE): 3E 40 00 00 05 02 01 00 04 00 7C

<< (CARD READER): 3E 00 00 00 00 3E

5. Then the CARD READER needs to send some data to the server. The CARD READER sends the subevent "0x03: SEND DATA".

<< (CARD READER): 3E 0F 00 00 C0 03 01 00 BC 60 03 33 00 00 08 00 20 38 00 00 00 C0 00 15 91 00 00 00 03 67 15 11 31 03 26 39 33 38 30 30 31 33 30 39 39 39 33 39 30 30 30 30 31 30 30 30 30 30 00 79 00 03 38 30 30 00 04 38 31

01 22 00 12 38 32 42 6C 75 65 50 61 64 2D 35 30 00 04 39 31 00 00 00 04 39 32 00 00 00 04 39 33 00 00 00 04 39 34 00 00 00 04 39 35 00 00 00 04 39 36 00 00 00 04 39 37 00 00 00 04 39 38 00 00 00 04 39 39 00 00 00 48 94 2D 02 E3 6F 16 09 0C 6D 9F B3 C7 6C 4C 83 03 61 11 03 1D 2A F5 FF 16 A2 7F BC AB 6B 5C E8 3E 4B 68 51 FD 77 AE A4 EF EF 8A 75 4C B3 A4 E6 DB 58 34 24 1A 00 00 00 00 36

6. The EXT DEVICE must confirm if it sent the data to the host successfully. It must send the subcommand "0x02: EVENT CONFIRM".

>> (EXT DEVICE): 3E 40 00 00 03 02 03 00 7C

<< (CARD READER): 3E 00 00 00 00 3E

7. Then the response from the host will be received from the EXT DEVICE and it needs to transfer the data to the CARD READER. The EXT DEVICE must call the subcommand "0x01: RECEIVE DATA".

>> (EXT DEVICE): 3E 40 00 00 70 01 00 6D 60 00 00 03 33 08 10 20 38 00 00 02 C0 00 05 91 00 00 00 03 67 15 11 52 03 26 30 30 39 33 38 30 30 31 33 30 39 39 39 33 39 30 30 30 30 31 30 30 30 30 30 00 48 94 2D 02 E3 6F 16 09 0C 6D 9F B3 C7 6C 4C 83 03 61 11 03 1D 2A F5 FF 16 A2 7F BC AB 6B 5C E8 3E 4B 68 51 FD 77 AE A4 EF EF 8A 75 4C B3 A4 E6 DB F5 7D E1 C3 00 00 00 00 B9

<< (CARD READER): 3E 00 00 00 00 3E

8. The CARD READER will send the subevent "0x02: SOCKET CLOSE".

<< (CARD READER): 3E 0F 00 00 02 02 01 30

9. The EXT DEVICE must confirm if it closed the socket successfully. It must send the subcommand "0x02: EVENT CONFIRM".

>> (EXT DEVICE): 3E 40 00 00 03 02 02 00 7D

<< (CARD READER): 3E 00 00 00 00 3E

10. The CARD READER sends the subevent "0x01: TRANSACTION COMPLETE (PAYMENT RESULT)".

<< (CARD READER): 3E 0E 00 00 1C 01 DF 05 04 00 00 00 00 DF 06 04 00 00 00 00 81 04 00 00 00 00 9F 41 04 00 00 03 67 15

11. The EXT DEVICE sends "0x02: GET RECEIPT TAGS" command to get information, which is needed to print a receipt for this transaction. In case of "Test Connection" there is no receipt, but the given example is from transaction "Purchase + TIP".

>> (EXT DEVICE): 3E 3D 00 00 5B 02 81 9F 04 9A 9F 21 9F 06 9F 26 9F 1C 9F 16 5F 2A 9F 41 5F 20 DF 00 DF 01 DF 02 DF 03 DF 04 DF 05 DF 06 DF 07 DF 08 DF 09 DF 0A DF 0B DF 10 DF 12 DF 19 DF 23 DF 25 DF 24 DF 26 DF 27 DF 28 DF 29 DF 2A DF 2B DF 2C DF 2D DF 2E DF 2F DF 30 DF 31 DF 60 DF 61 DF 62 DF 63 DF 64 18

<< (CARD READER): 3E 00 00 01 42 81 04 00 00 00 1E 9A 03 19 03 20 9F 21 03 18 22 22 9F 06 07 A0 00 00 00 03 20 10 9F 26 08 5D 58 40 B4 E2 51 EE 9B 9F 1C 08 39 33 38 30 30 31 33 30 9F 16 0F 39 39 39 33 39 30 30 30 30 31 30 30 30 30 30 30 5F 2A 02 09 75 9F 41 04 00 00 03 65 5F 20 10 58 58 58 58 58 58 58 20 20 58 58 58 58 58 58 58 DF 00 0D 56 69 73 61 20 45 6C 65 63 74 72 6F 6E DF 04 04 00 00 00 0F DF 05 04 00 00 00 00 DF 06 04 00 00 00 00 DF 07 0C 30 30 30 30 30 36 36 39 31 31 33 35 DF 08 06 39 36 39 33 36 34 DF 09 04 00 00 00 00 DF 0A 10 2A 2A 2A 2A 2A 2A 2A 2A 2A 2A 2A 2A 37 38 39 38 DF 10 01 01 DF 12 02 00 00 DF 23 02 00 00 DF 25 02 00 01 DF 26 02 0A 02 DF 27 03 42 47 4E DF 28 07 39 37 30 32 36 31 30 DF 29 04 31 30 30 30 DF 2A 04 54 45 53 54 DF 2E 06 44 41 54 45 43 53 DF 2F 19 42 55 4C 2E 54 5A 41 52 49 47 52 41 44 53 4B 4F 20 53 48 41 4F 53 53 45 45 DF 30 05 53 4F 46 49 41 DF 31 04 54 45 53 54 DF 60 04 00 00 00 03 DF 61 02 00 27 DF 62 02 43 31 DF 63 04 00 00 00 0A DF 64 04 00 00 00 05 CC

12. The EXT DEVICE sends "0x03: TRANSACTION END" command.

>> (EXT DEVICE): 3E 3D 00 00 01 03 01

<< (CARD READER): 3E 00 00 00 00 3E

Note: If the CARD READER has capability for communicating with the host by its own, then the steps 3-9 will be missing from the transaction flow.

12. CARD READER ERROR STRINGS

These are some of the more common error codes, which can be returned in tag 0xDF06. Please check “6. Display result message” what should be the logic for displaying the transaction result string.

DF06 values:

1 (0x01): “General error”

2 (0x02): “Invalid command”

3 (0x03): “Invalid parameter”

5 (0x05): “Invalid length”

7 (0x07): “Operation is not permitted”

8 (0x08): “No data”

9 (0x09): “Timeout”

12 (0x0C): “Invalid device”

18 (0x12): “Operation canceled”

21 (0x15): “Wrong password”

31 (0x1F): “Operation is not permitted”

50 (0x32): “No connection with the host”

51 (0x33): “Please, use chip”

52 (0x34): “Please, end day”

13. HOST ERROR STRINGS

These are some of the more common error codes, which can be returned in tag 0xDF09. Please check “6. Display result message” what should be the logic for displaying the transaction result string.

DF09 values:

00 (0x00): “Approved transaction”

04 (0x04): “Pick up the card”

06 (0x06): “Technical problem”

07 (0x07): “Pick up the card”

12 (0x0C): “Invalid transaction”

13 (0x0D): “Invalid amount”

14 (0x0E): “Invalid card number”

15 (0x0F): “Unable to route to issuer host”

33 (0x21): “Expired card”

36 (0x24): “Restricted card. Pick up the card”

37 (0x25): “Pick up the card and call security dept.”

38 (0x26): “Allowable PIN tries exceeded”

41 (0x29): “Lost card. Pick up the card”

43 (0x2B): “Stolen card. Pick up the card”

51 (0x33): “Insufficient funds”

52 (0x34): “No checking account”

53 (0x35): “No savings account”

54 (0x36): “Expired card”

55 (0x37): “Incorrect PIN”

57 (0x39): “Transaction is not permitted to the cardholder”

58 (0x3A): “Transaction is not permitted to the terminal”

65 (0x41): “Exceeded withdrawal frequency limit”

66 (0x42): “Pick up the card and call security dept.”

67 (0x43): “Pick up the card and call security dept.”

75 (0x4B): “Allowable number of PIN tries exceeded”

78 (0x4E): “Account balance unavailable”

79 (0x4F): “Unacceptable PIN”

82 (0x52): “Timeout”

86 (0x56): “PIN validation not possible”

87 (0x57): “Partial approved transaction”

91 (0x5B): “Issuer is inoperative”

92 (0x5C): “Issuer is inoperative”

94 (0x5E): “Duplicated transaction”

96 (0x60): “System malfunction”

14. 0x0B: EMV subevents

1. 0x82: USER INTERFACE

DESCRIPTION:

At some point during the transaction this event can be sent from the CARD READER. This event will be sent only from the device BlueCash-50! In this case, if the [MESSAGE ID: 2] value is any of the described below ones, then the EXT DEVICE should display the message. If the [MESSAGE ID: 2] is missing in the below examples, then it can be ignored.

CARD READER:

0B 00 LH LL [SUBEVENT] 00 C1 02 [MESSAGE ID: 2]

[MESSAGE ID: 2]:

- 00 10: „Remove card”
- 00 15: “Present card”
- 00 16: “Processing”
- 00 17: “Card read OK. Please remove card.”
- 00 18: “Try other interface”
- 00 1B: “Online authorization”
- 00 1C: “Try other card”
- 00 1D: “Insert card”
- 00 20: “See phone”

00 21: “Present card again”
00 F0: „Please use the chip reader”
00 F1: “Please insert, read or try another card”

EXAMPLE:

(CARD READER -> EXT DEVICE) >> 3E 0B 00 00 06 82 00 C1 02 00 F0 82

CLARIFICATION:

(CARD READER -> EXT DEVICE):

'>':	0x3E - Start packet.
EVENT:	0x0B - EMV event.
00:	0x00 - Fixed value '00'.
LH LL:	0x00 0x06 - The data length is 6 bytes.
SUBEVENT:	0x82 – USER INTERFACE subevent.
00:	0x00 - Fixed value '00'.
C1:	0xC1 - Fixed value ‘C1’.
02:	0x02 - Fixed tag length '02'.
[MESSAGE ID: 2]:	0x00 0xF0 - „Please use the chip reader”
CSUM:	0x82 - Xor of all bytes (3E 0B 00 00 06 82 00 C1 02 00 F0).